

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

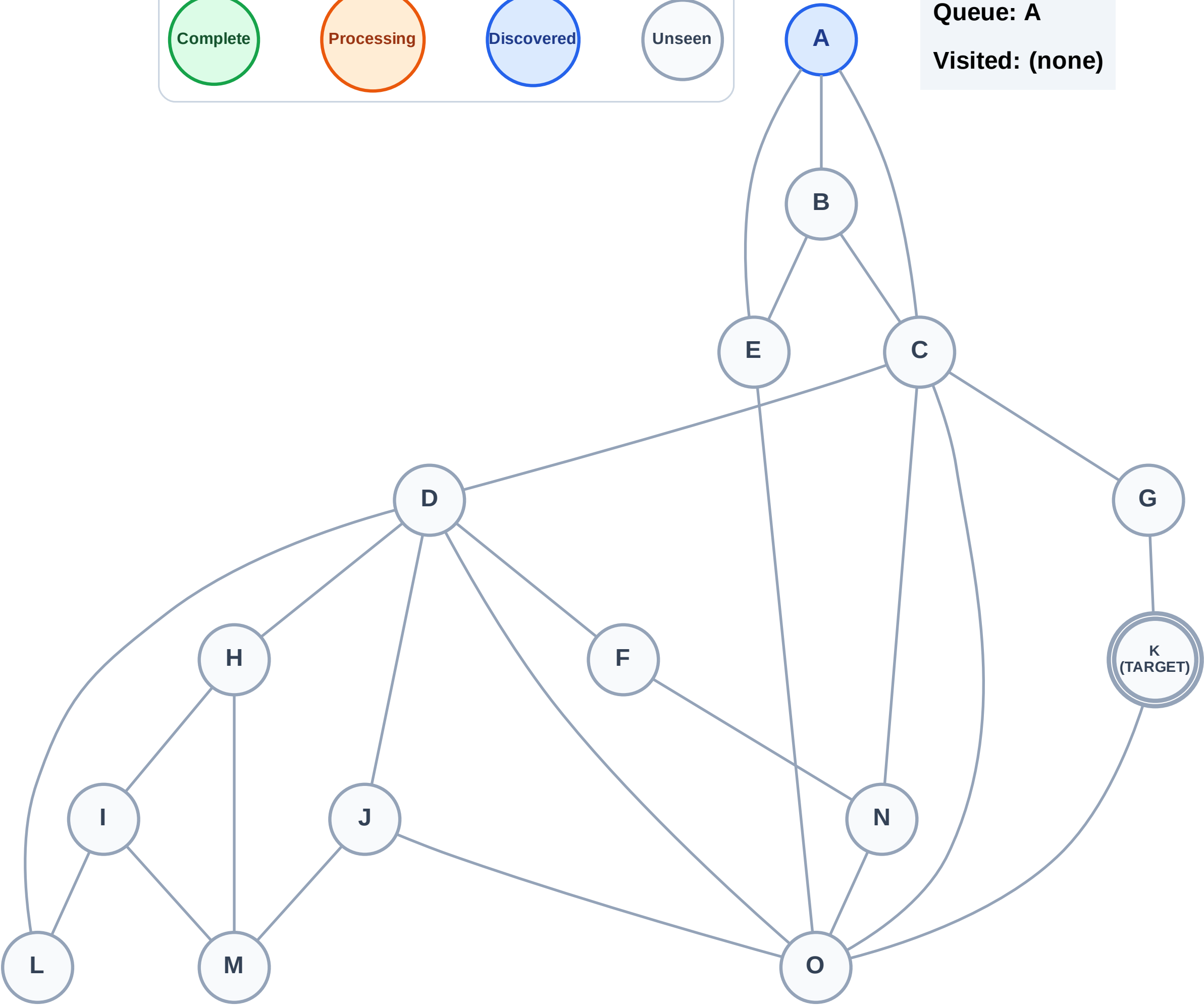
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



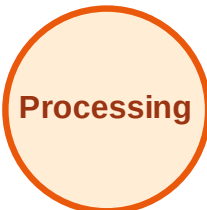
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



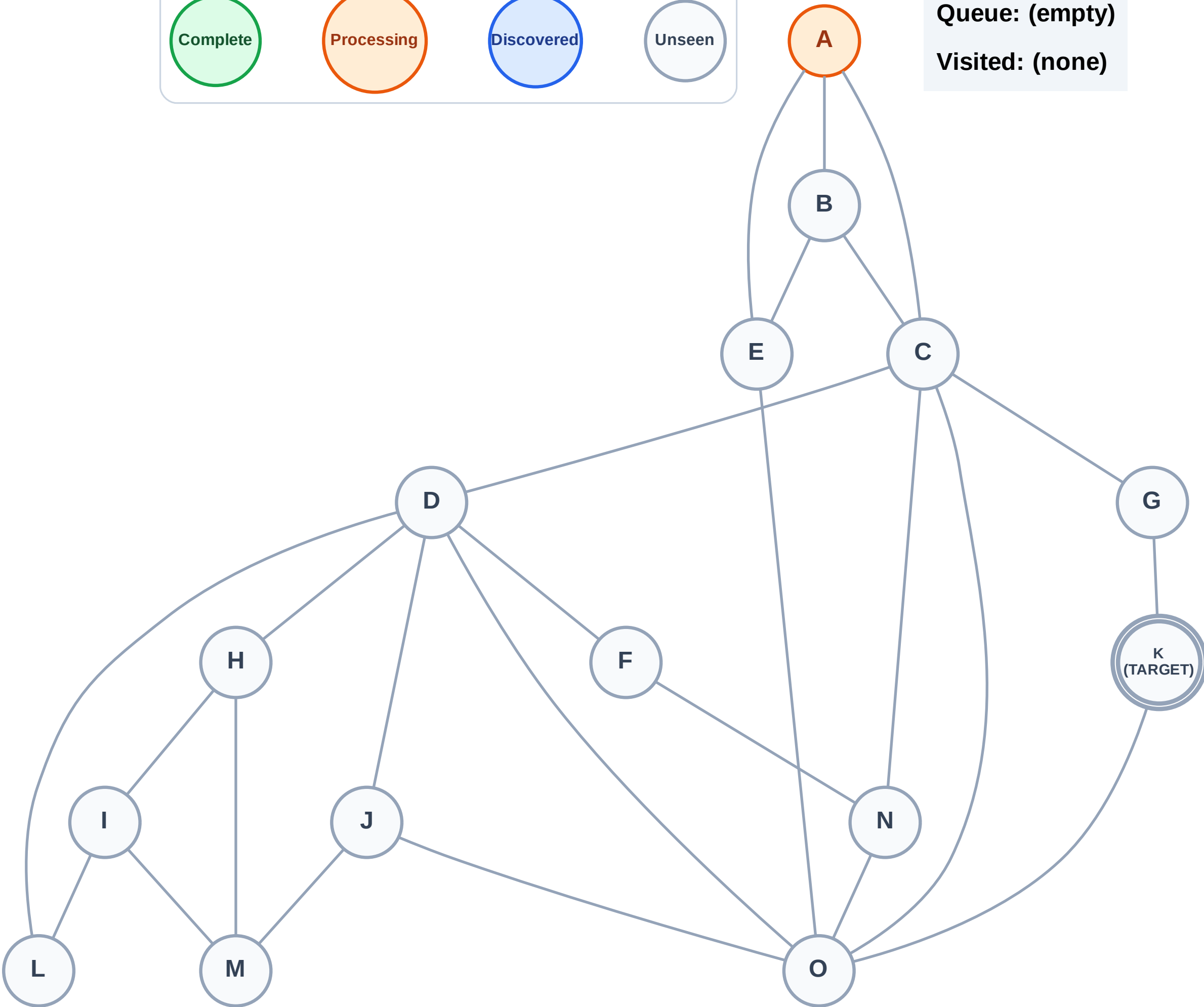
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

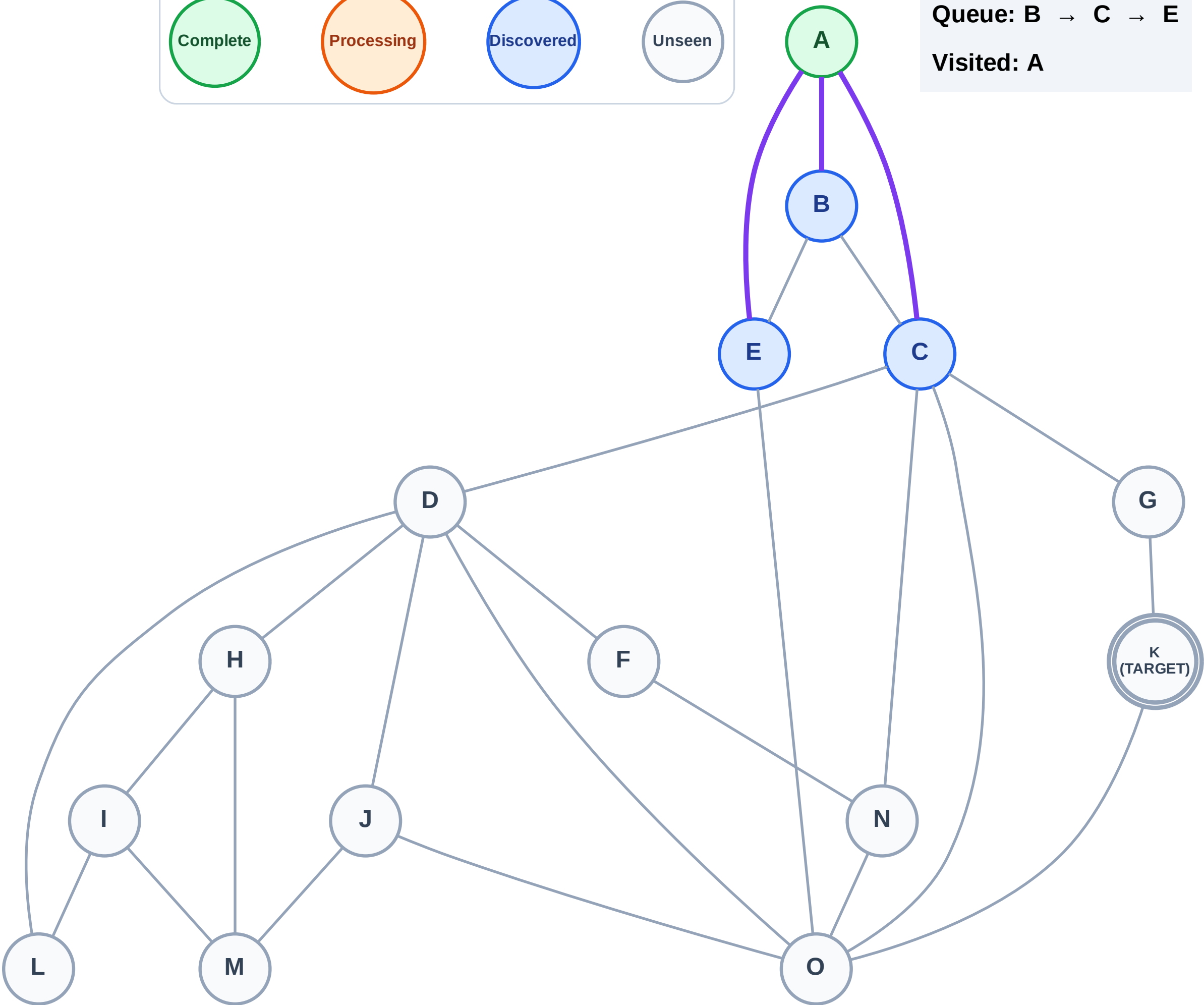
Step 3: Discover B, C, E from A

Legend



Queue: B → C → E

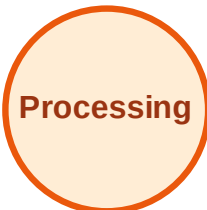
Visited: A



BFS Traversal

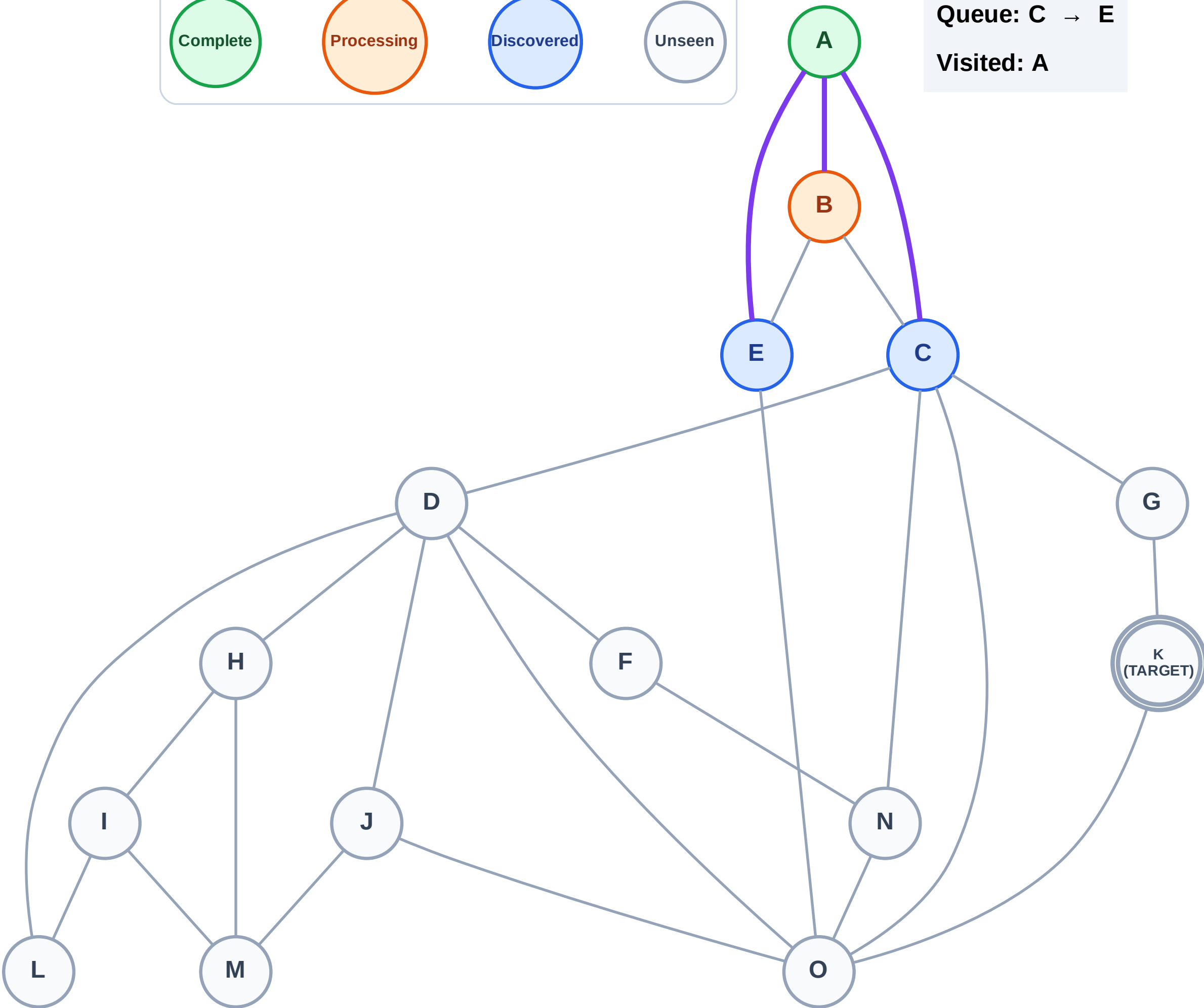
Step 4: Process node B

Legend



Queue: C → E

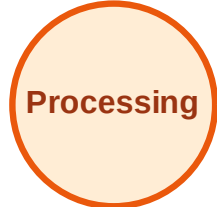
Visited: A



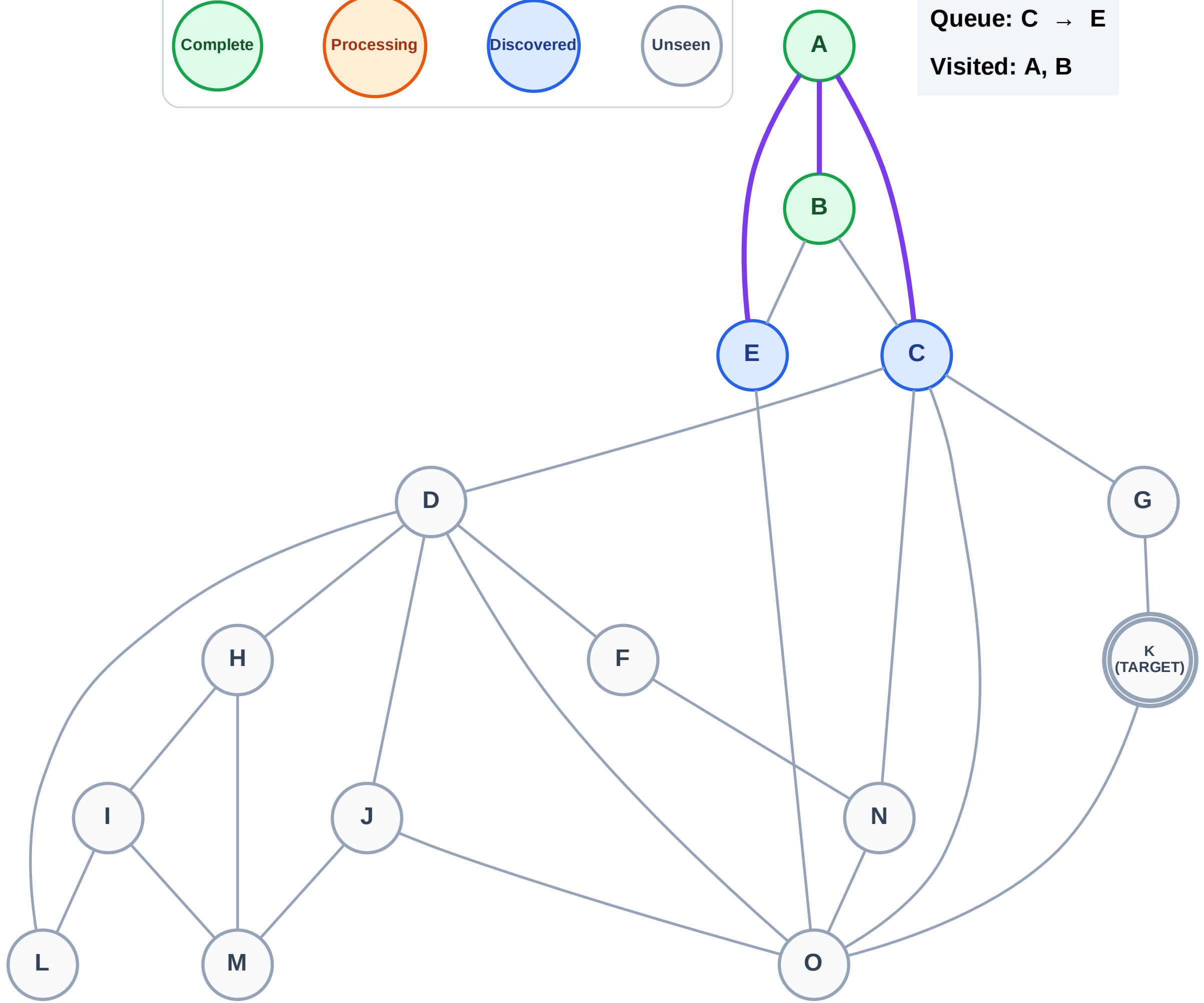
BFS Traversal

Step 5: No new neighbors from B

Legend



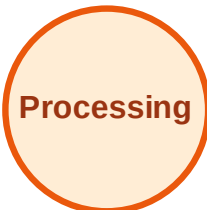
Queue: C → E
Visited: A, B



BFS Traversal

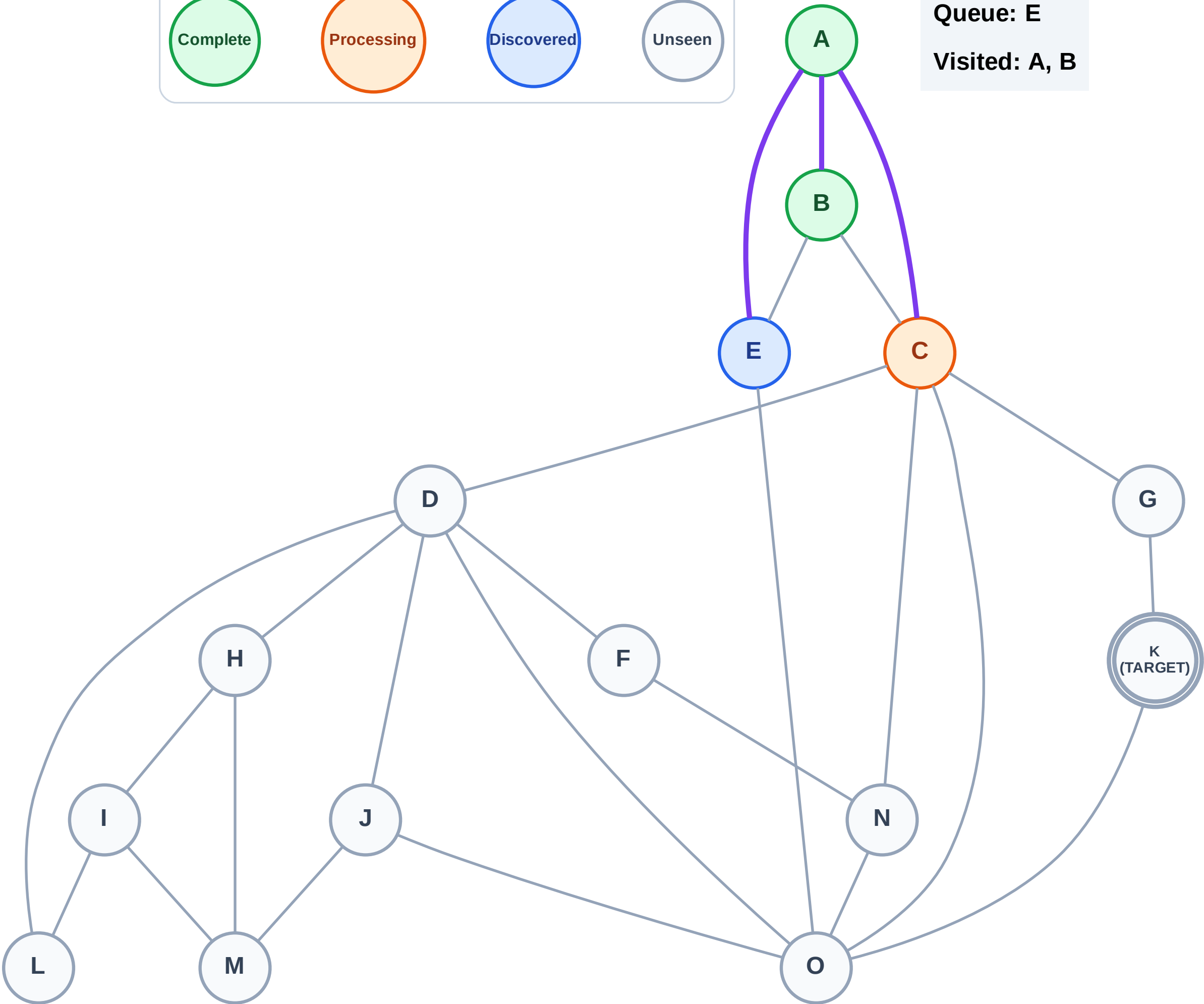
Step 6: Process node C

Legend



Queue: E

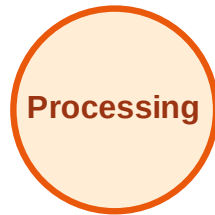
Visited: A, B



BFS Traversal

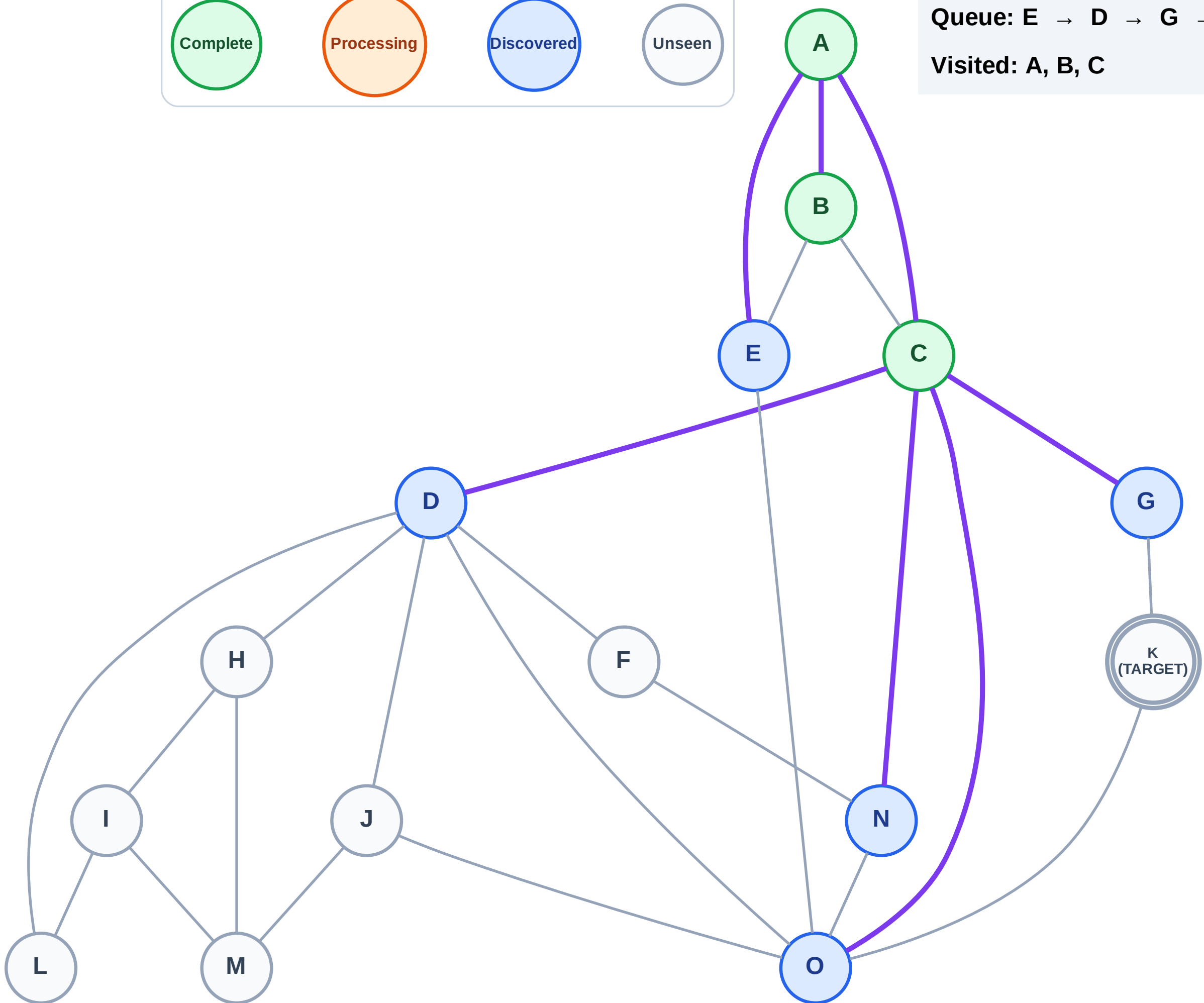
Step 7: Discover D, G, N, O from C

Legend

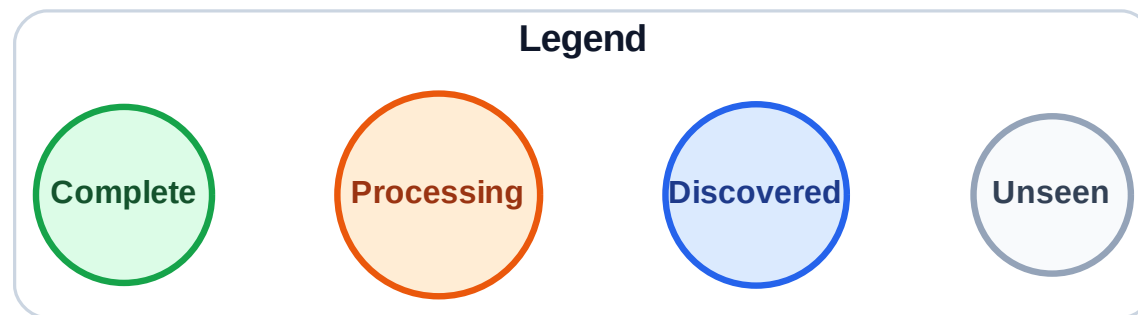


Queue: E → D → G → N → O

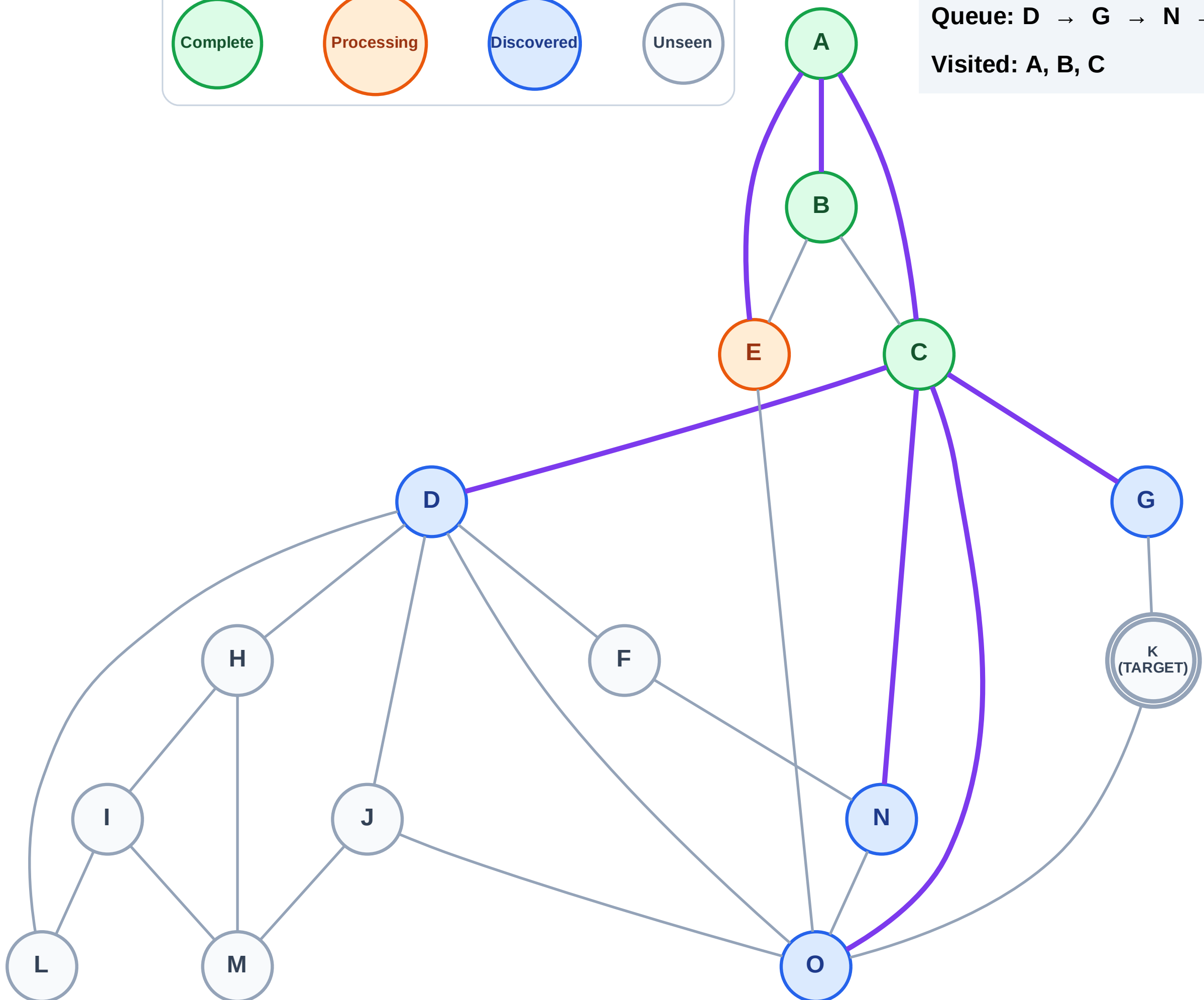
Visited: A, B, C



Step 8: Process node E



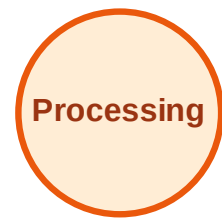
Visited: A, B, C



BFS Traversal

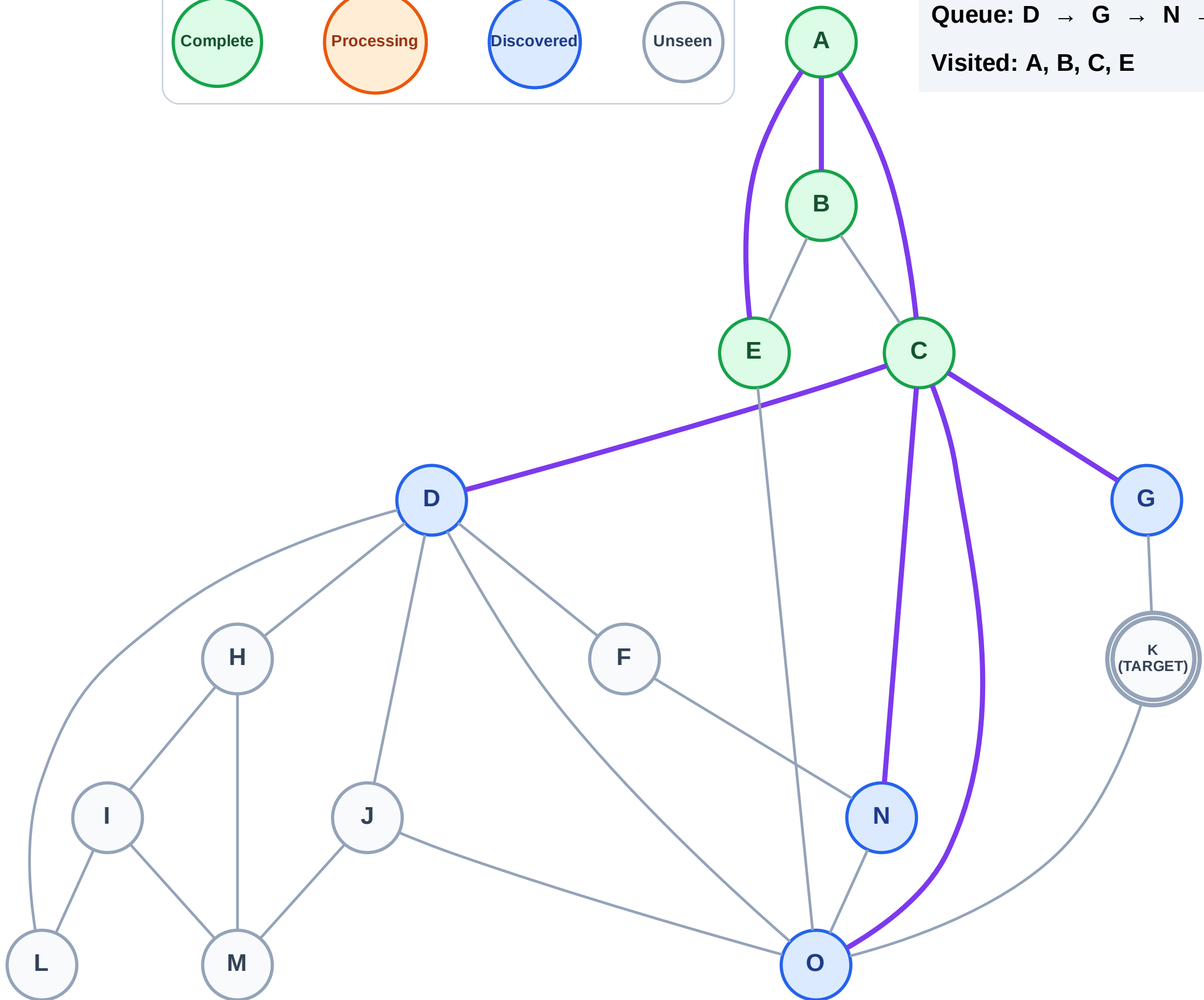
Step 9: No new neighbors from E

Legend



Queue: D → G → N → O

Visited: A, B, C, E



BFS Traversal

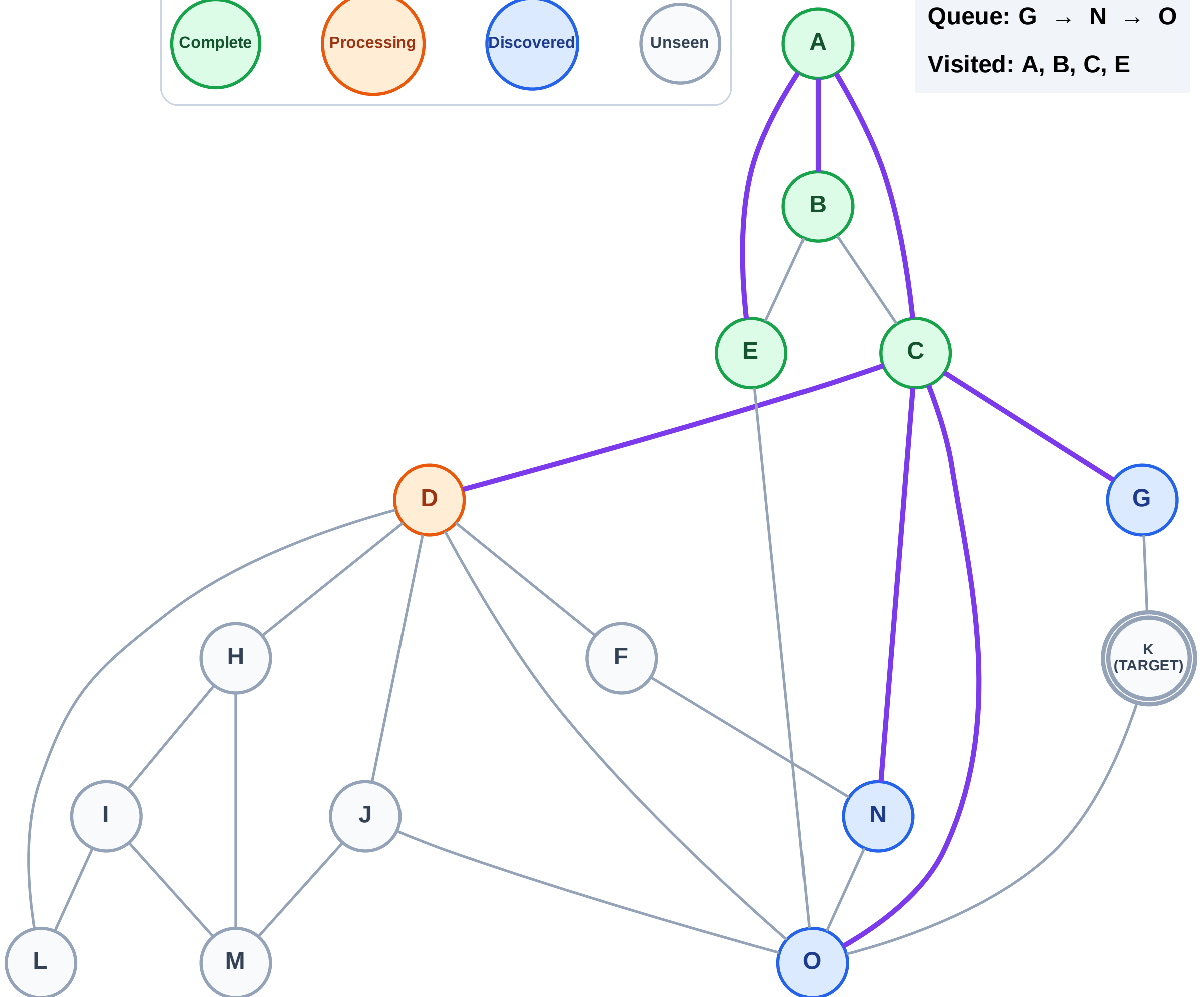
Step 10: Process node D

Legend



Queue: G → N → O

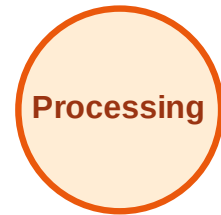
Visited: A, B, C, E



BFS Traversal

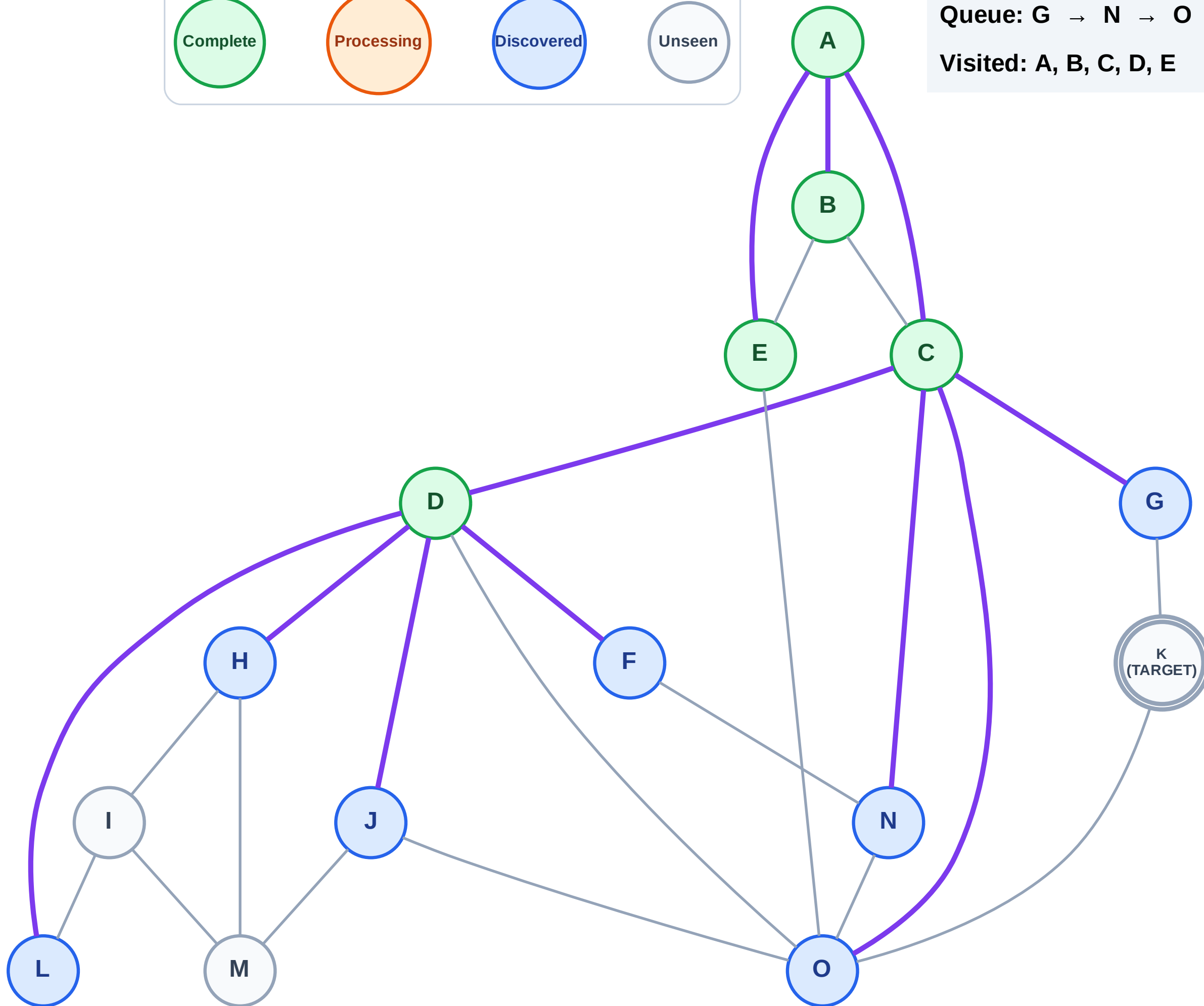
Step 11: Discover F, H, J, L from D

Legend



Queue: G → N → O → F → H → J → L

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node G

Legend

Complete

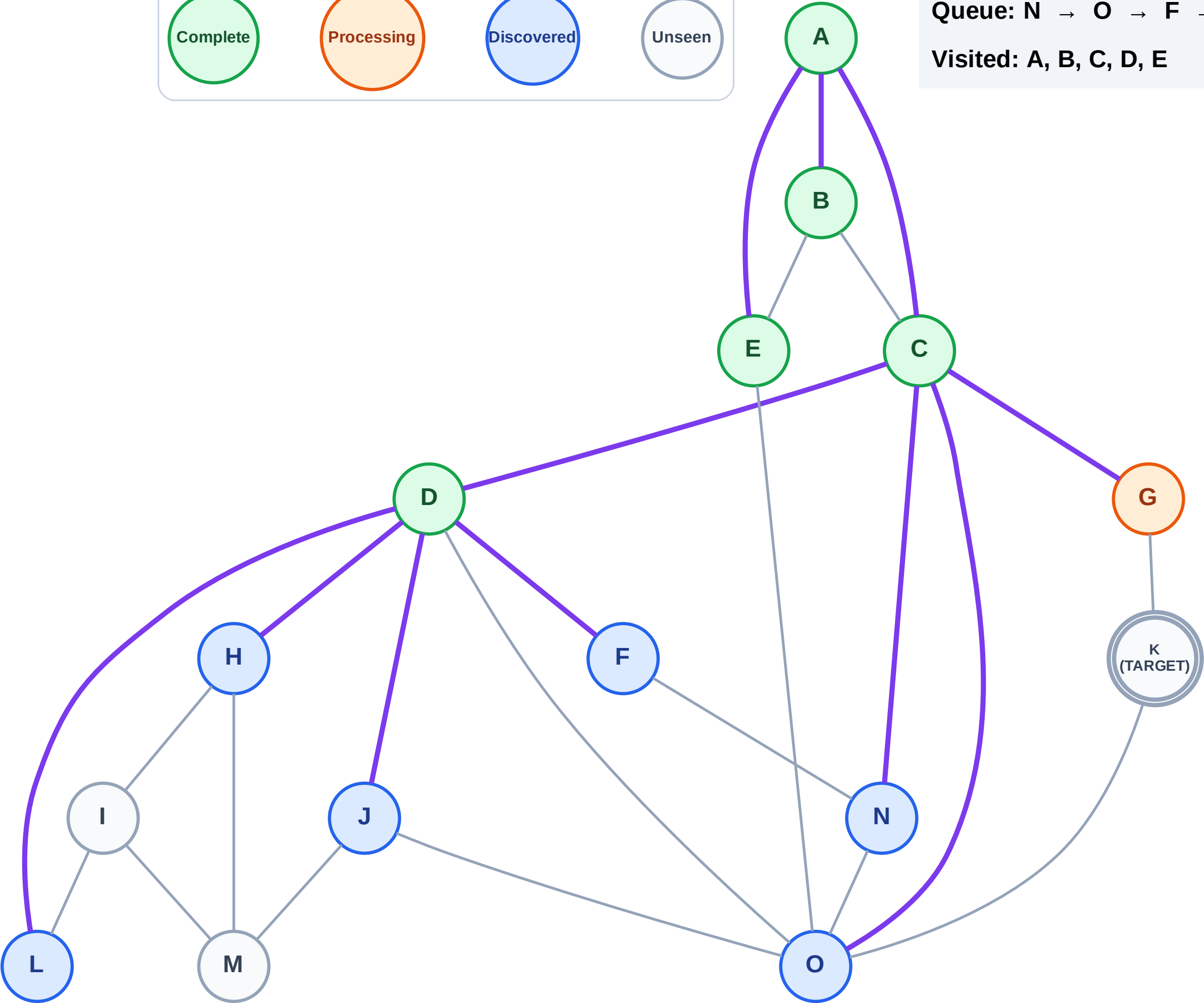
Processing

Discovered

Unseen

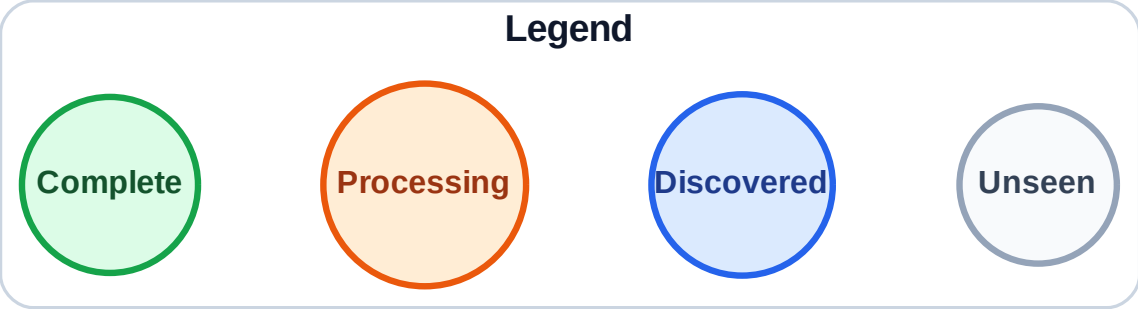
Queue: N → O → F → H → J → L

Visited: A, B, C, D, E



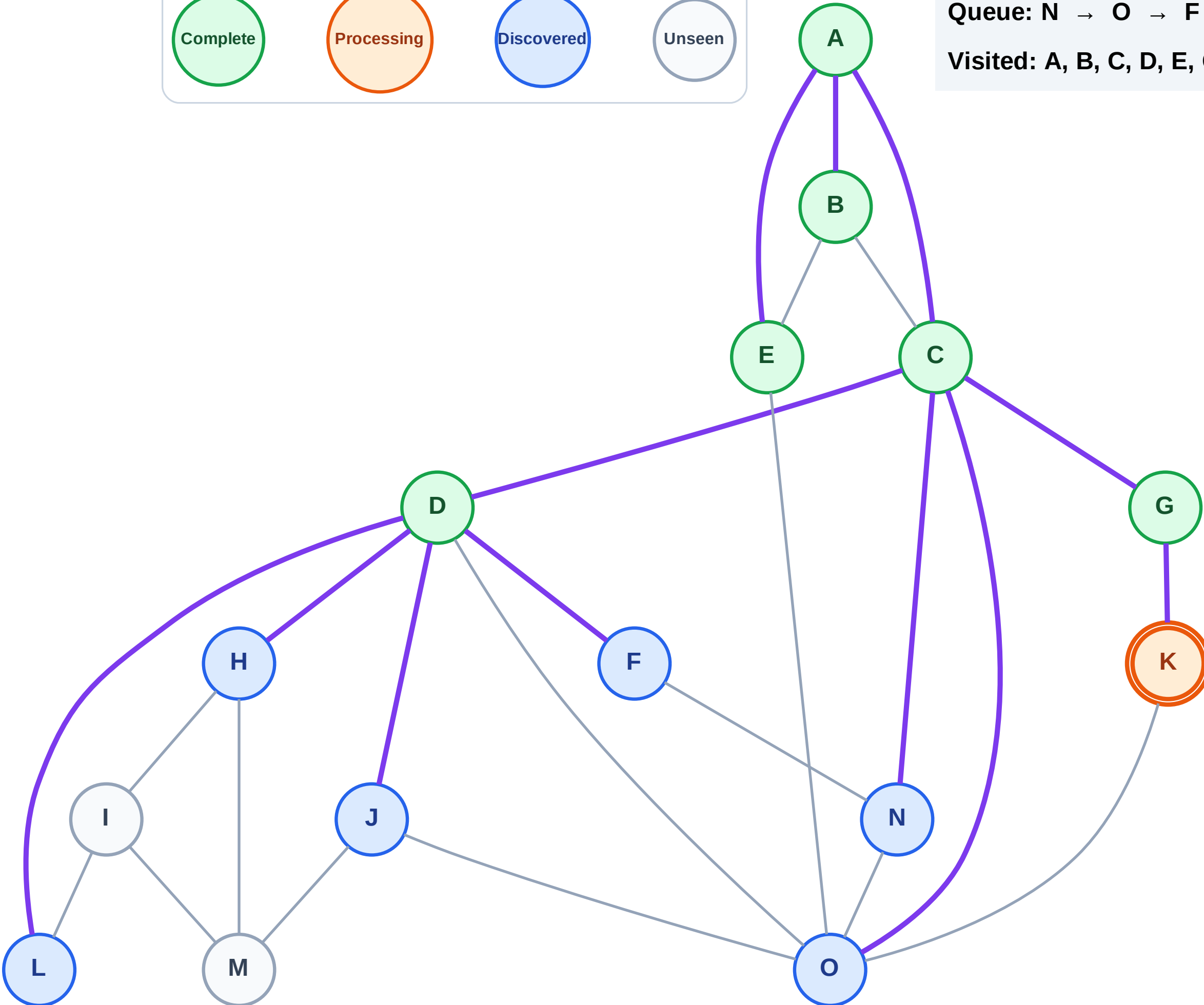
BFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): $A \rightarrow C \rightarrow G \rightarrow K$



Queue: N → O → F → H → J → L

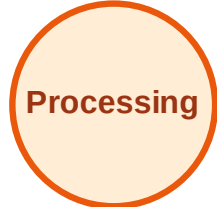
Visited: A, B, C, D, E, G, K



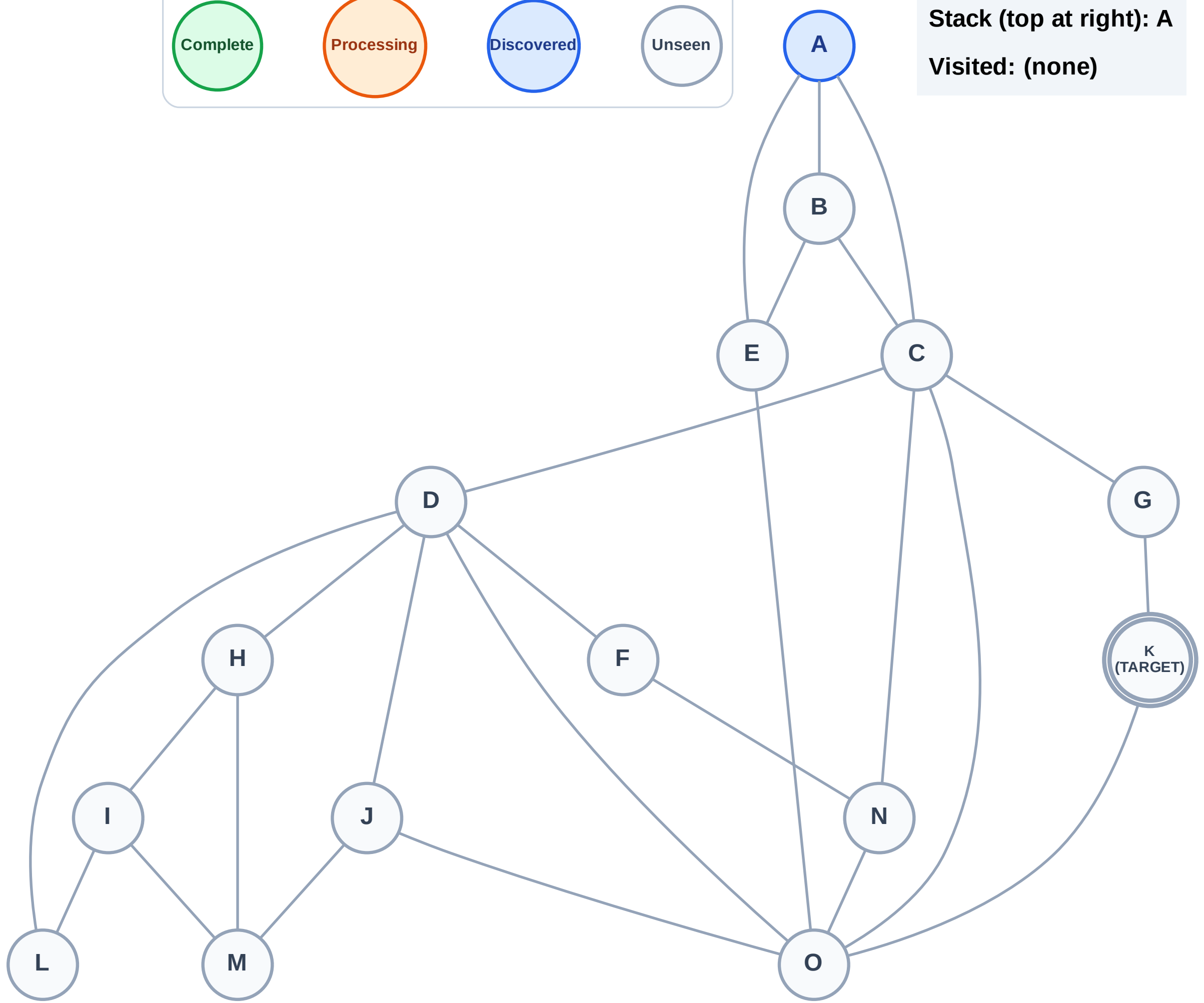
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



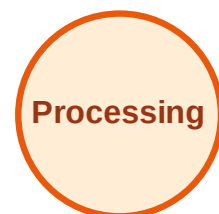
Stack (top at right): A
Visited: (none)



DFS Traversal

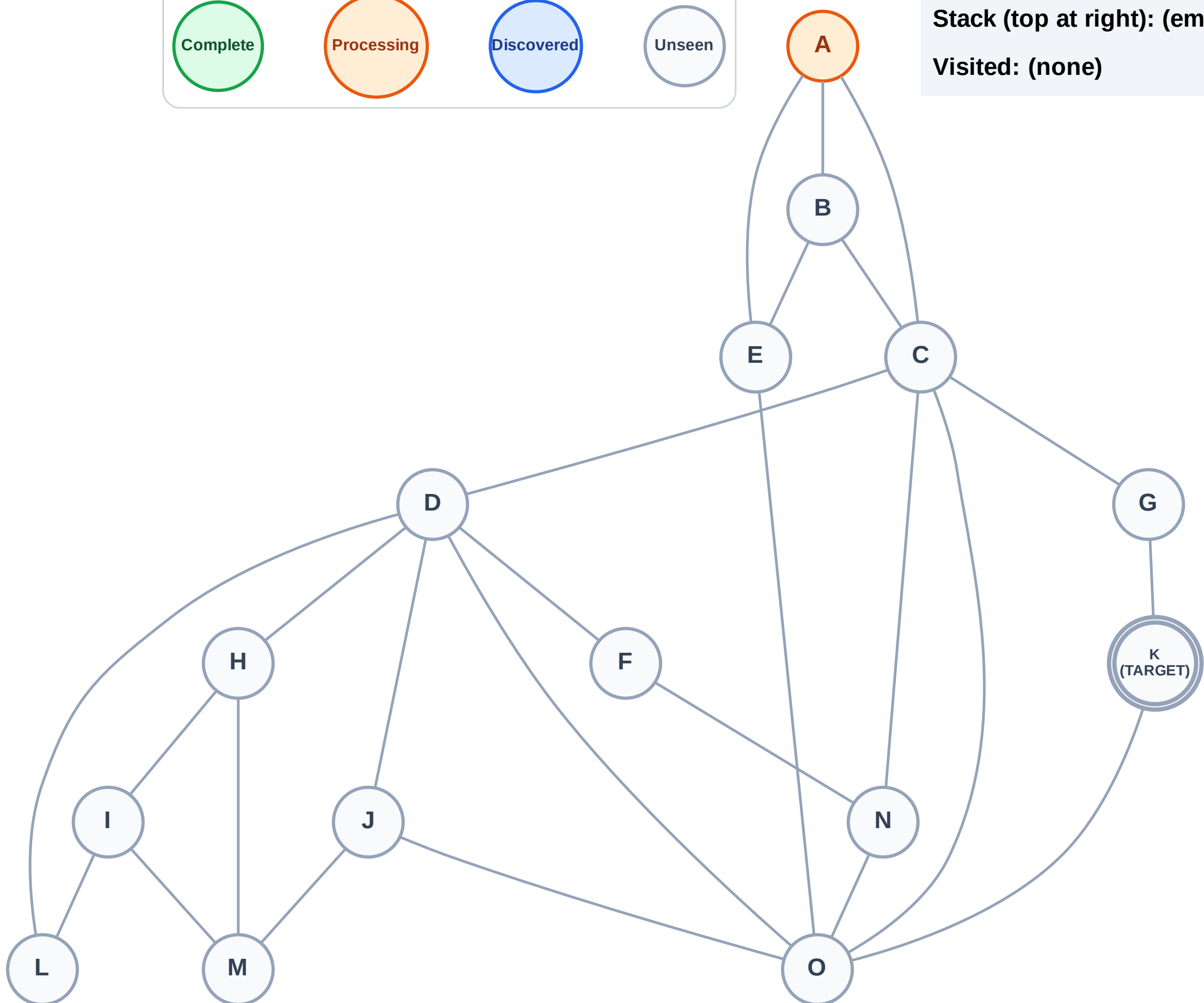
Step 2: Process node A

Legend



Stack (top at right): (empty)

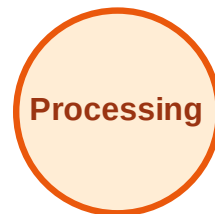
Visited: (none)



DFS Traversal

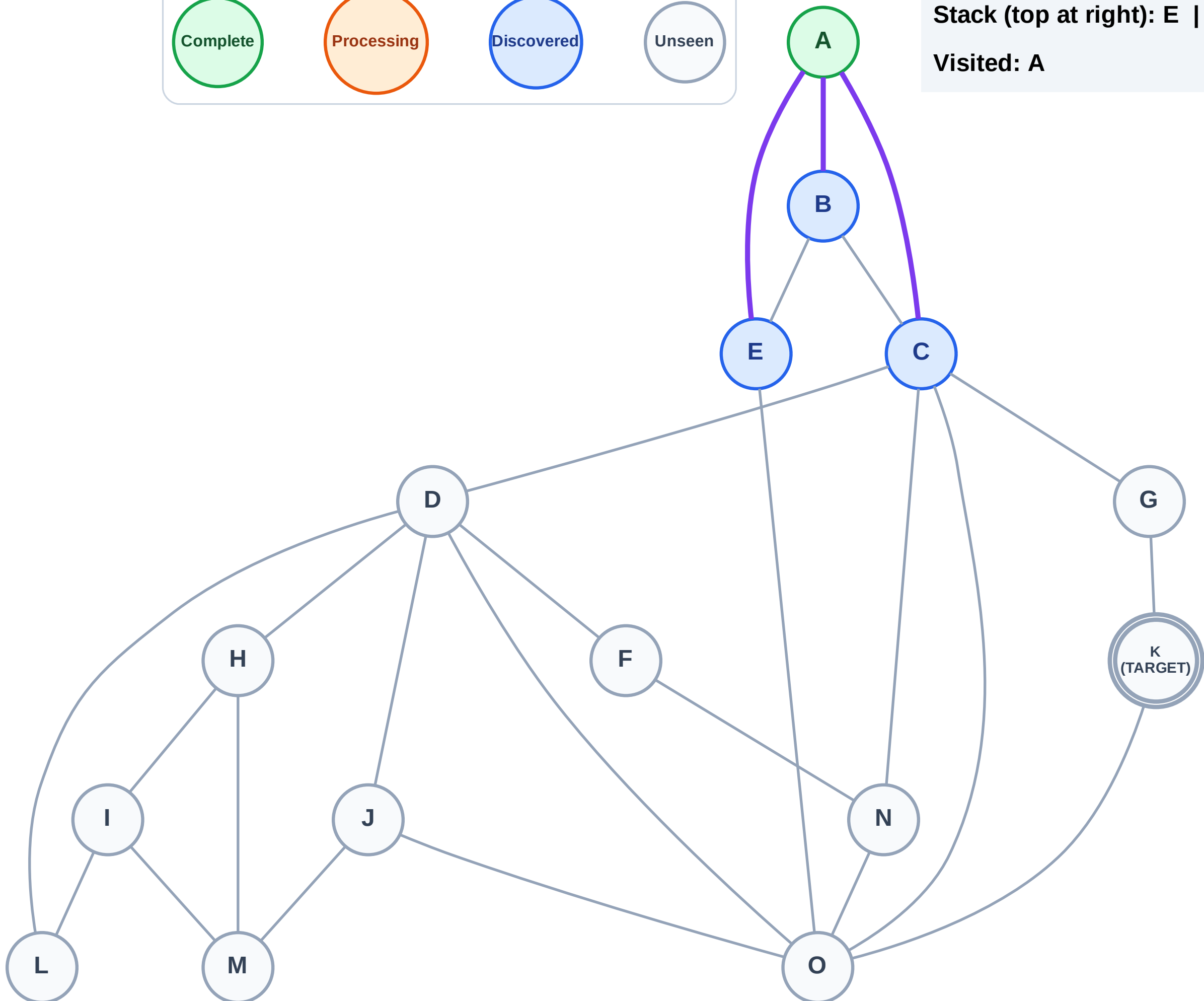
Step 3: Discover E, C, B from A

Legend



Stack (top at right): E | C | B

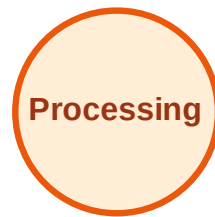
Visited: A



DFS Traversal

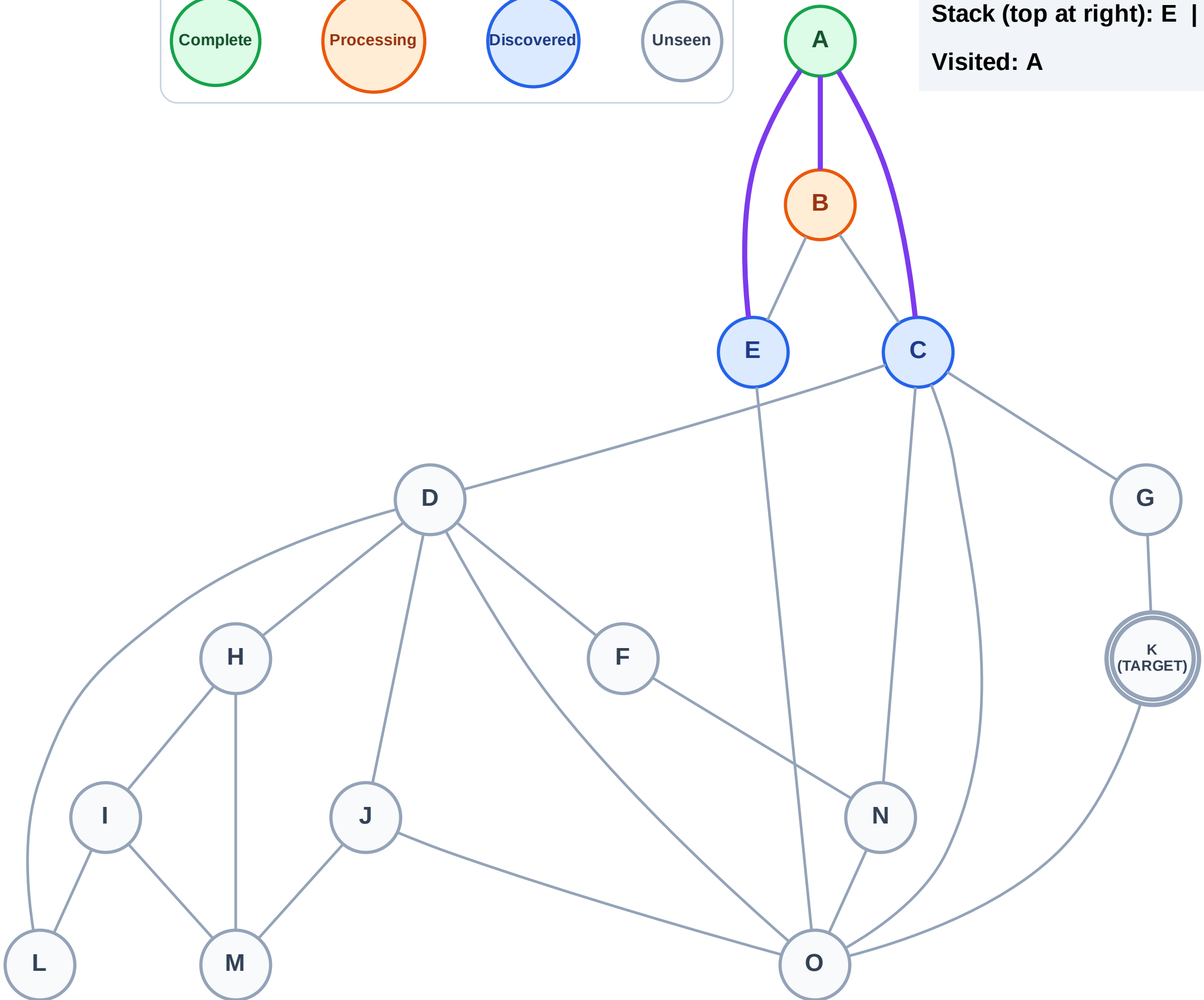
Step 4: Process node B

Legend



Stack (top at right): E | C

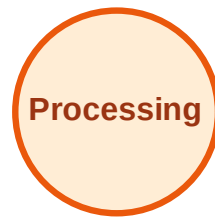
Visited: A



DFS Traversal

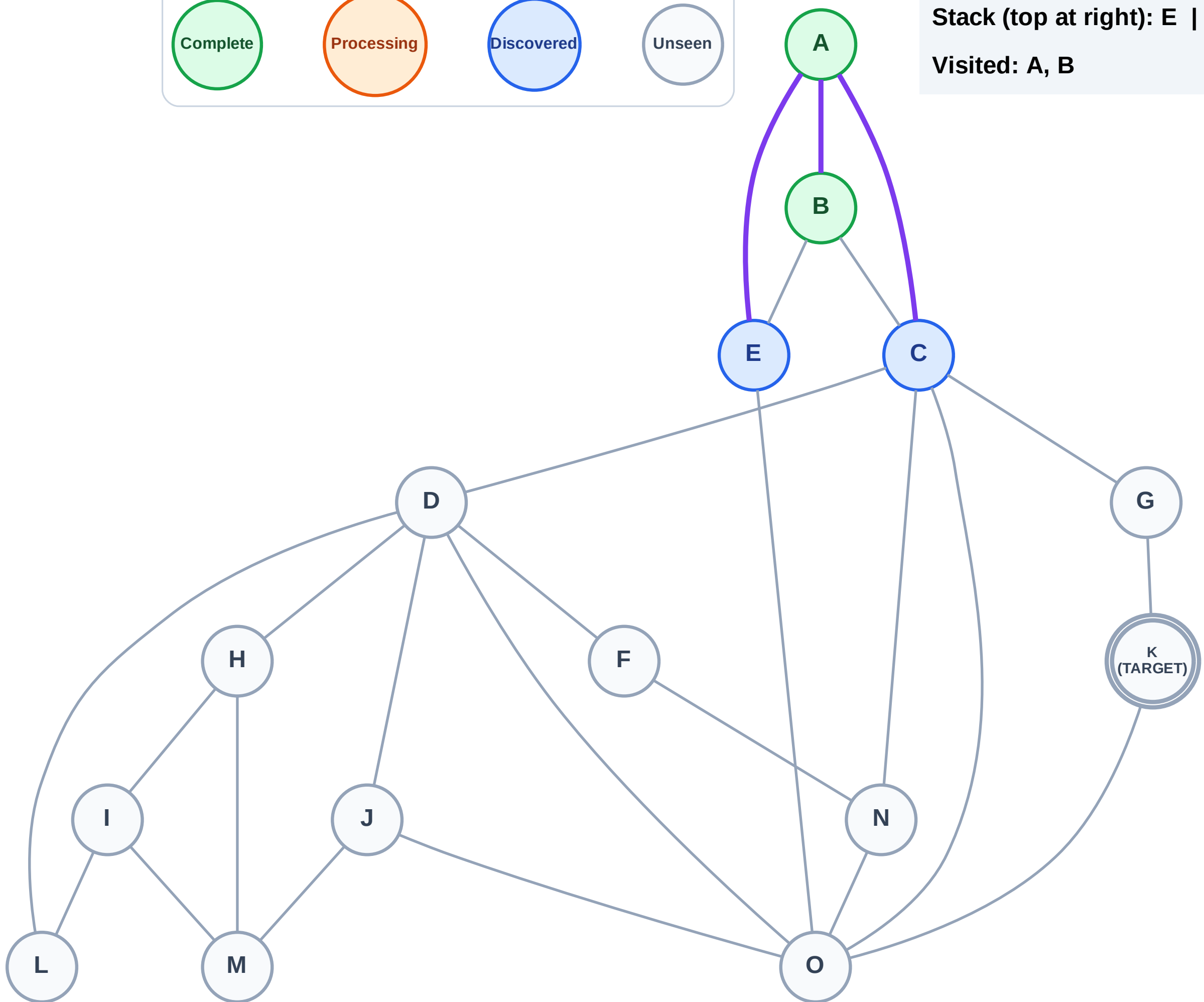
Step 5: No new neighbors from B

Legend



Stack (top at right): E | C

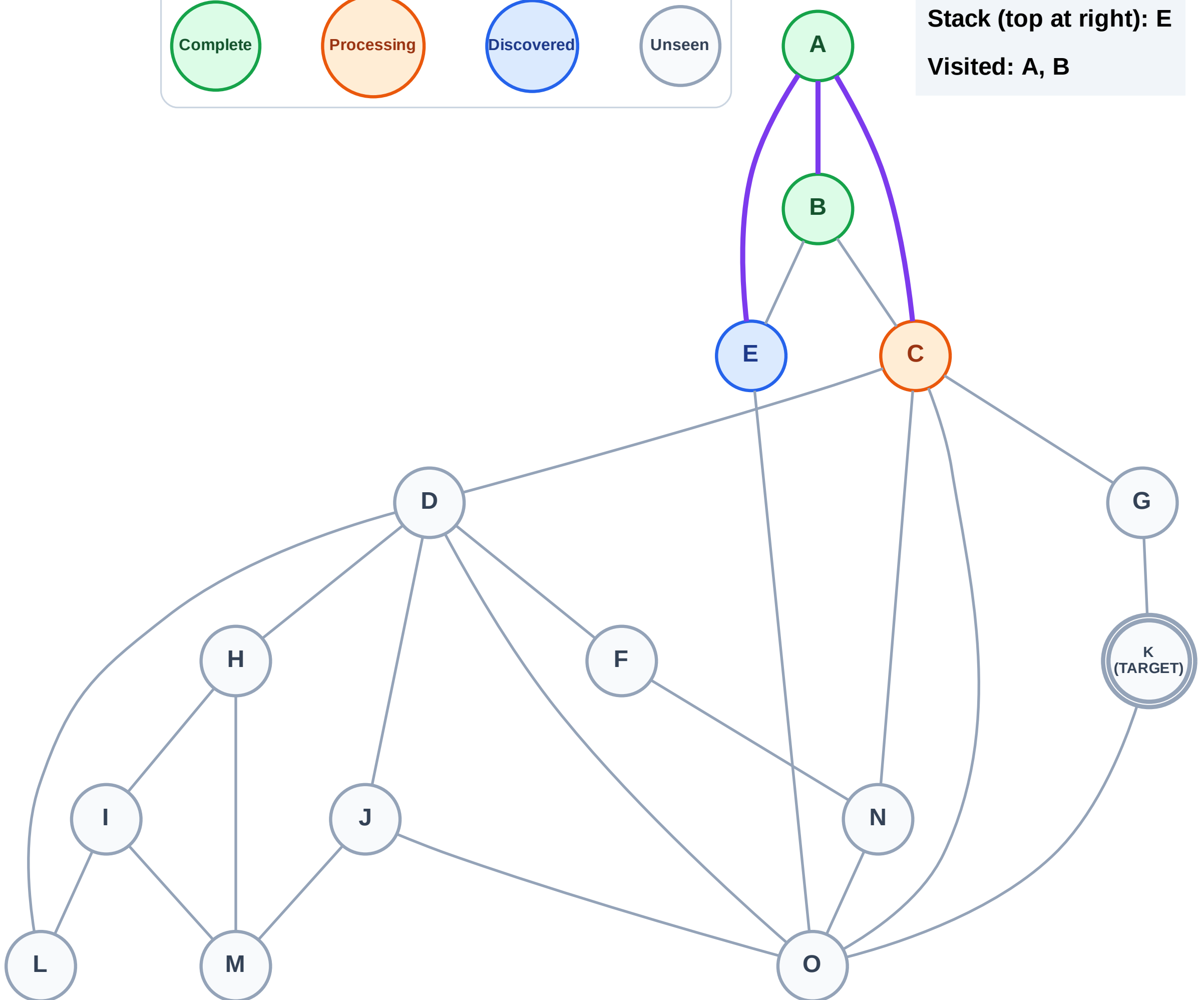
Visited: A, B



Step 6: Process node C



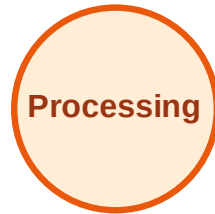
Visited: A B



DFS Traversal

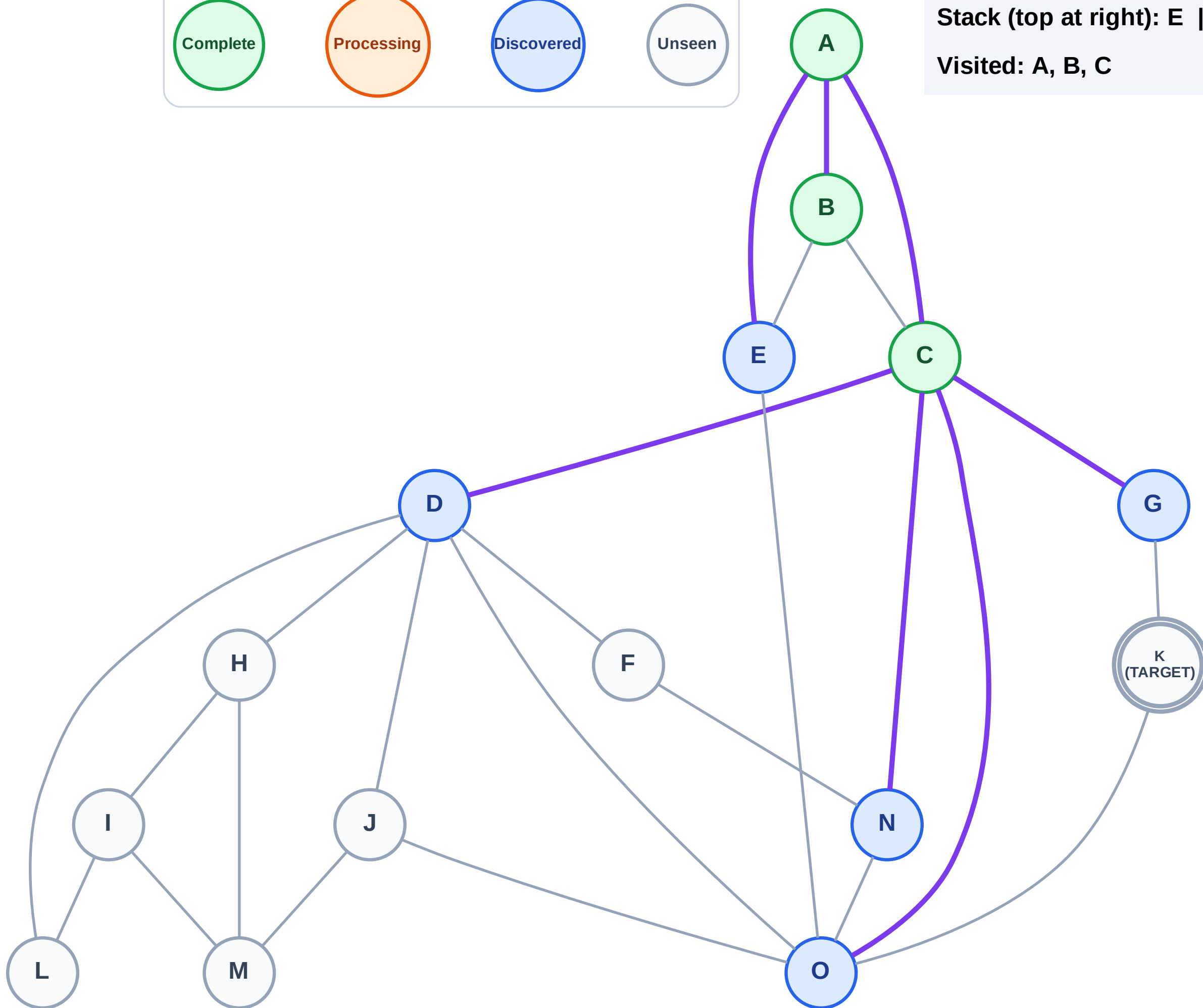
Step 7: Discover O, N, G, D from C

Legend



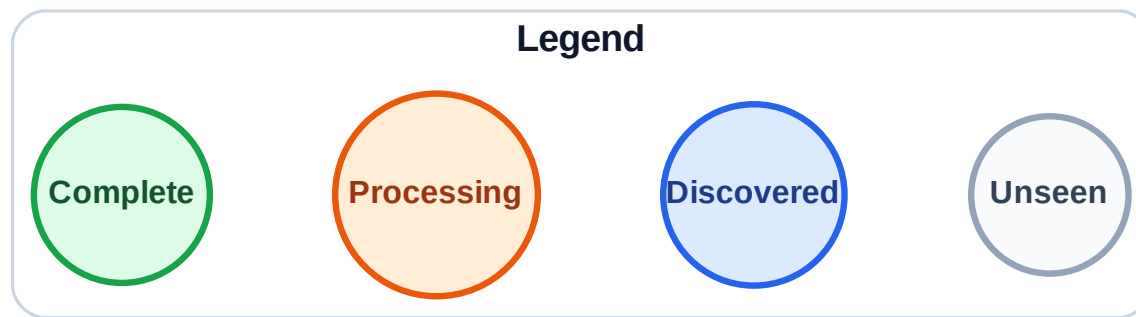
Stack (top at right): E | O | N | G | D

Visited: A, B, C



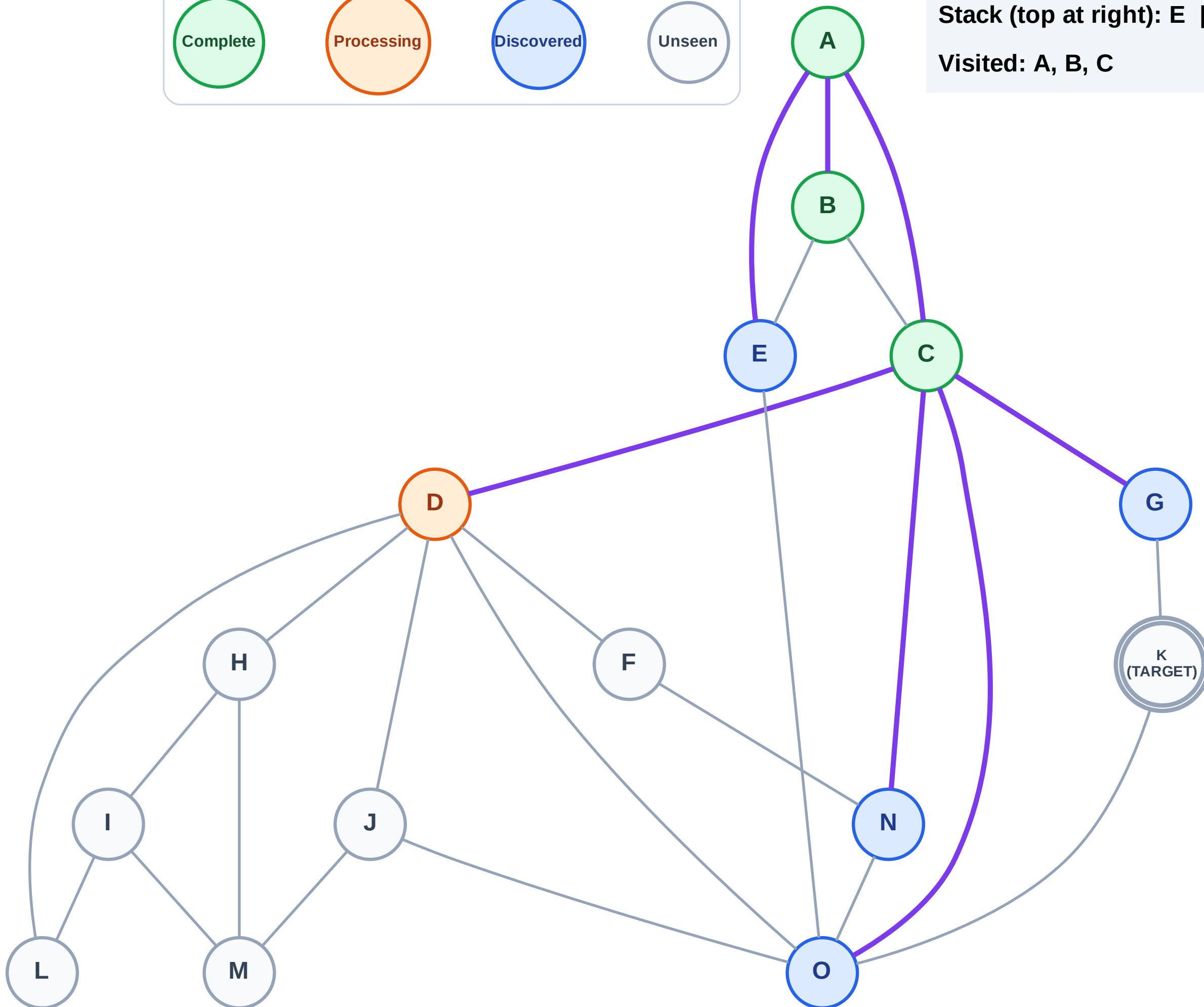
Step 8: Process node D

Step 8: Process node D



Stack (top at right): E | O | N | G

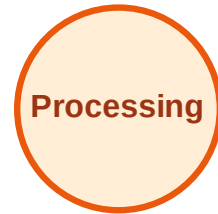
Visited: A, B, C



DFS Traversal

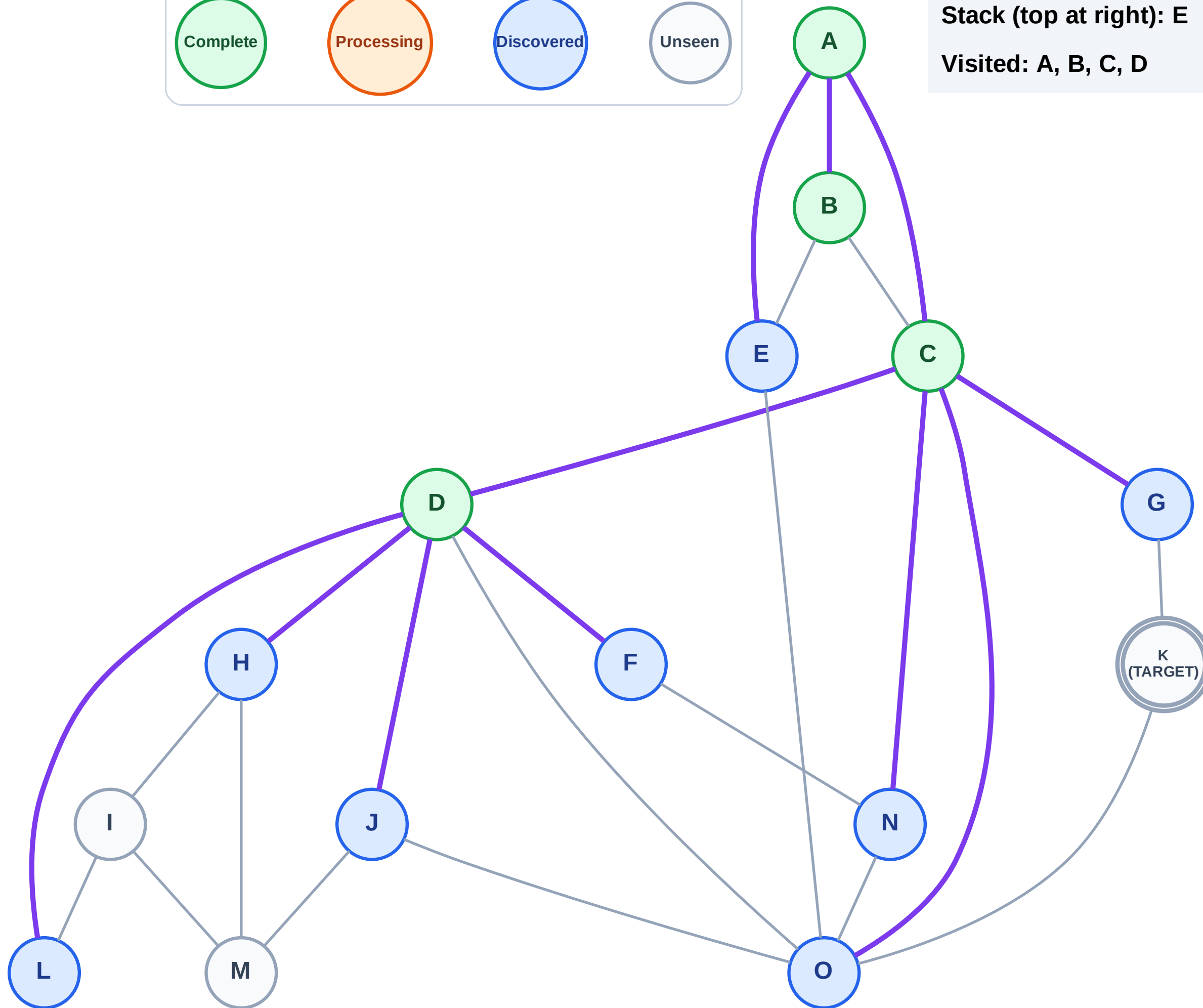
Step 9: Discover L, J, H, F from D

Legend

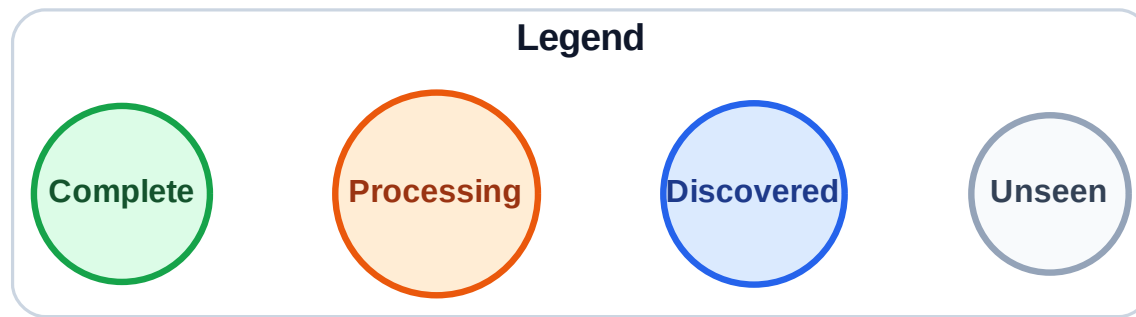


Stack (top at right): E | O | N | G | L | J | H | F

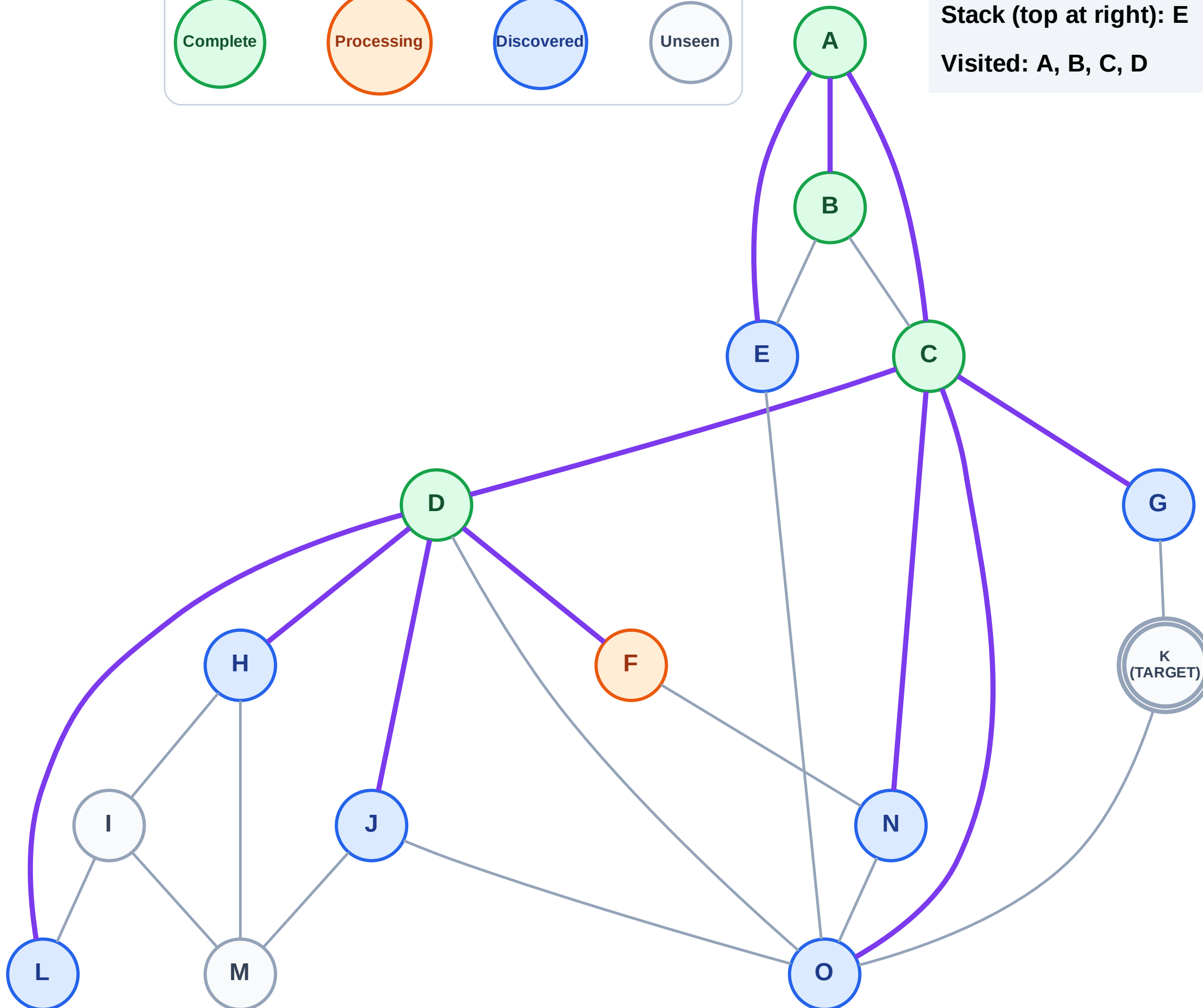
Visited: A, B, C, D



Step 10: Process node F



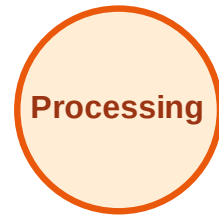
Visited: A, B, C, D



DFS Traversal

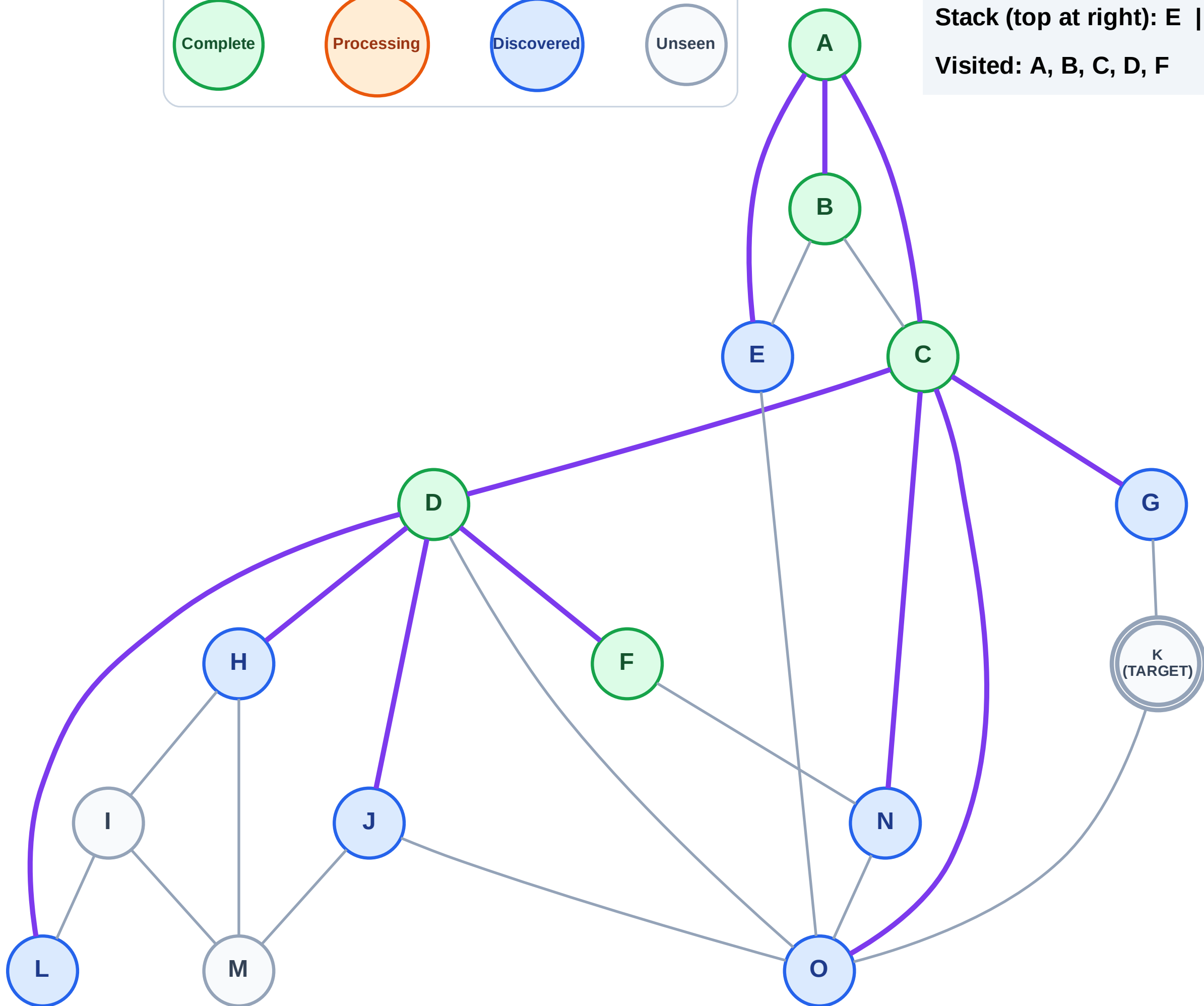
Step 11: No new neighbors from F

Legend



Stack (top at right): E | O | N | G | L | J | H

Visited: A, B, C, D, F



DFS Traversal

Step 12: Process node H

Legend

Complete

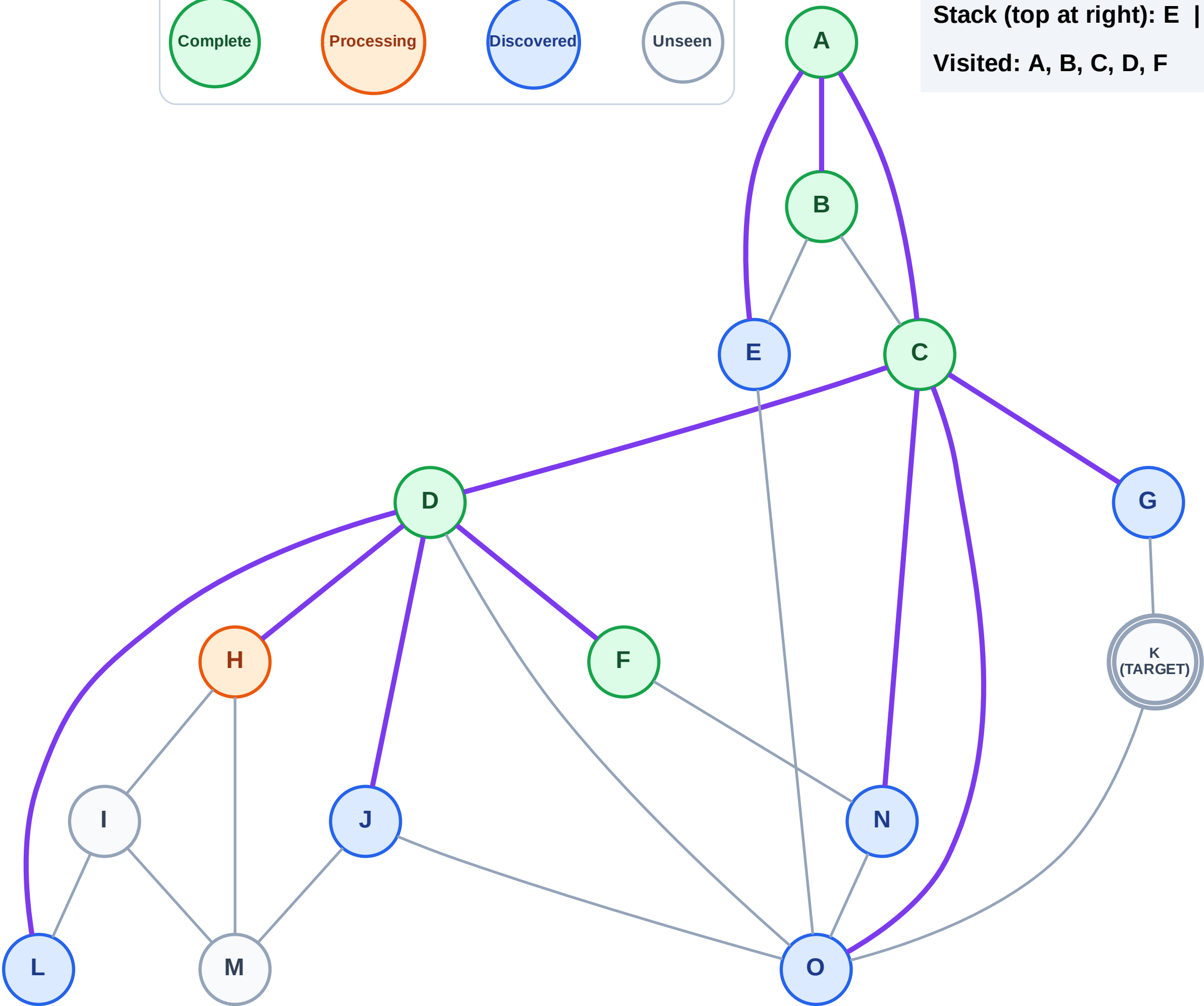
Processing

Discovered

Unseen

Stack (top at right): E | O | N | G | L | J

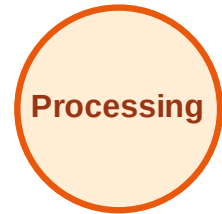
Visited: A, B, C, D, F



DFS Traversal

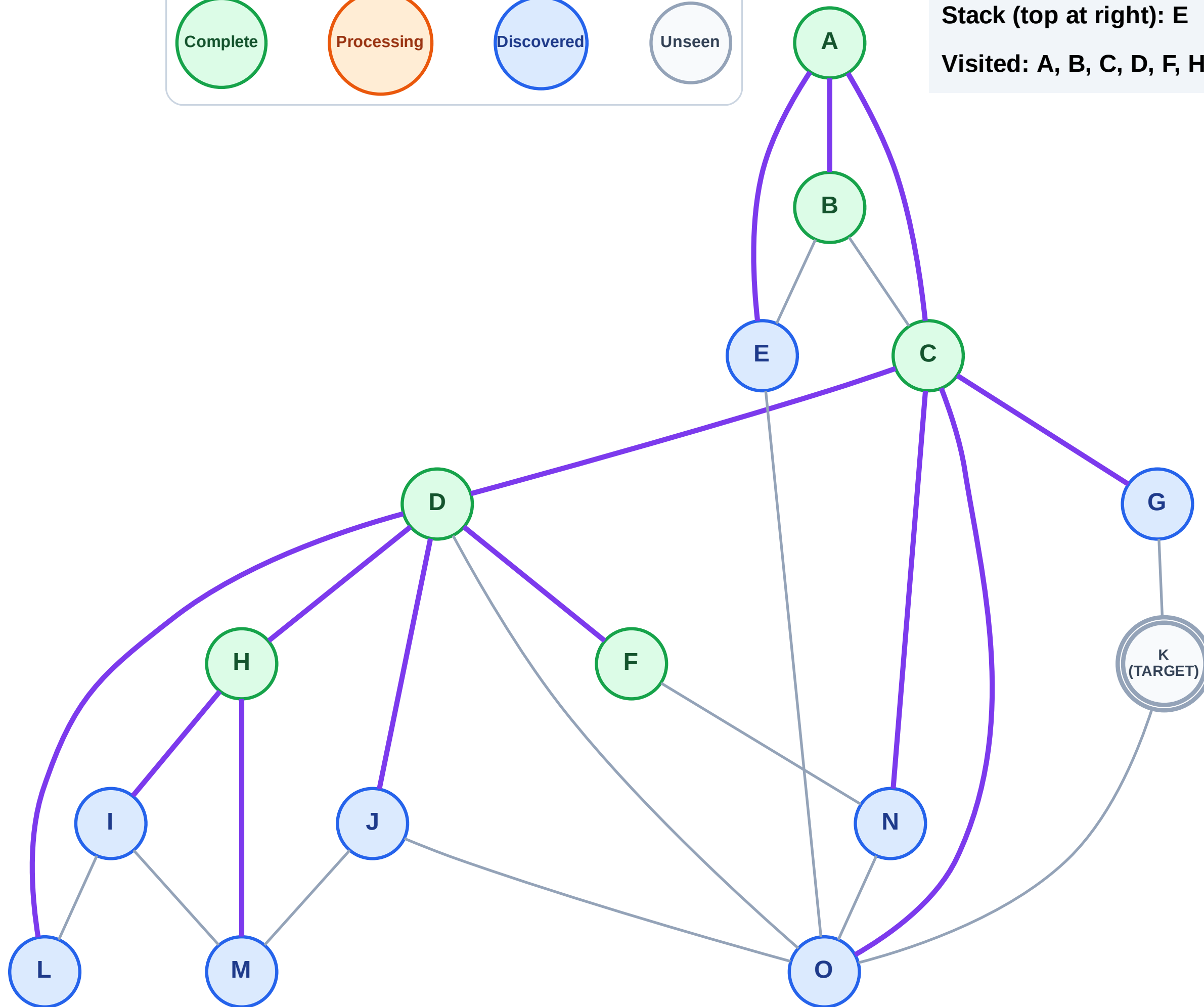
Step 13: Discover M, I from H

Legend



Stack (top at right): E | O | N | G | L | J | M | I

Visited: A, B, C, D, F, H



DFS Traversal

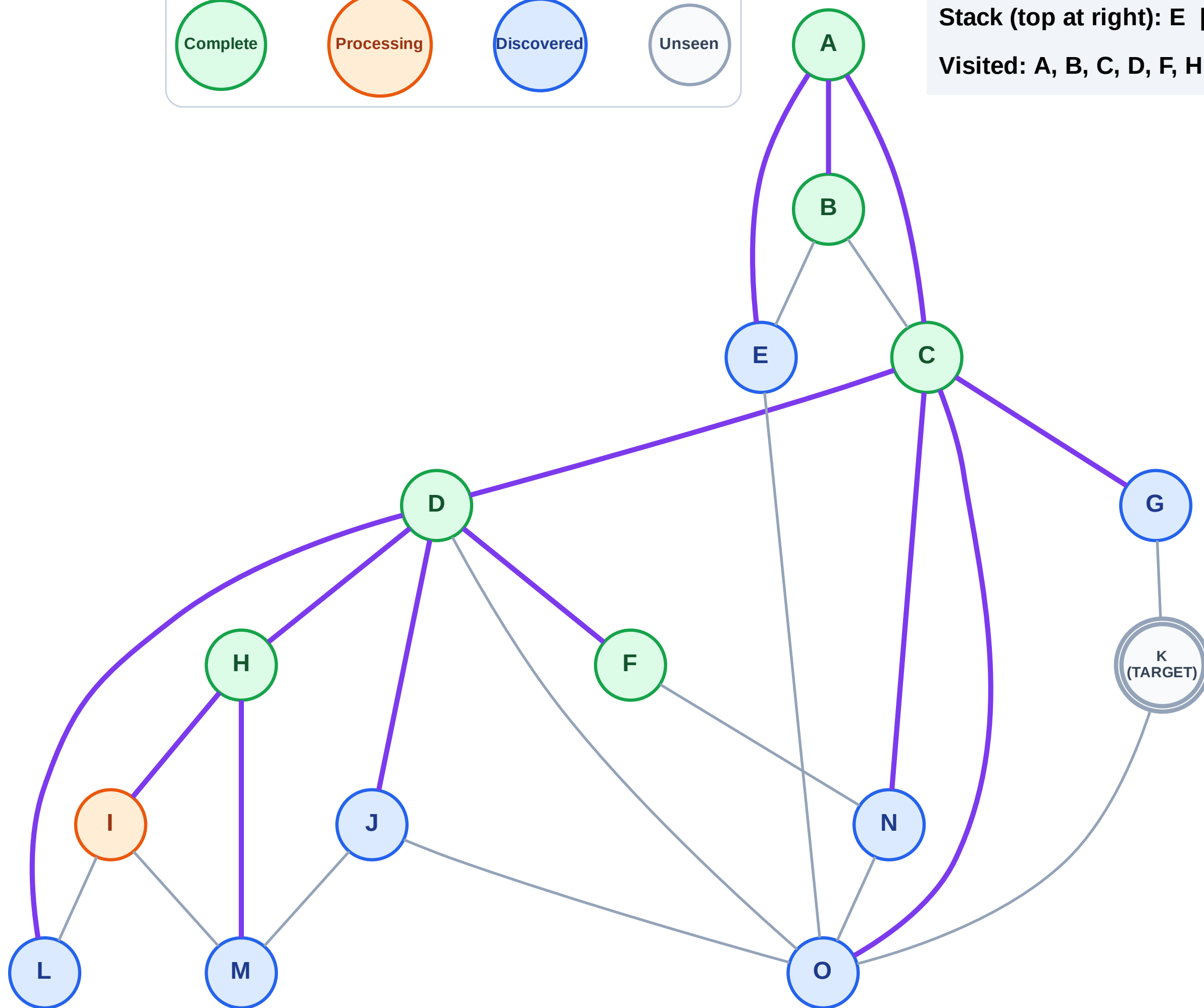
Step 14: Process node I

Legend



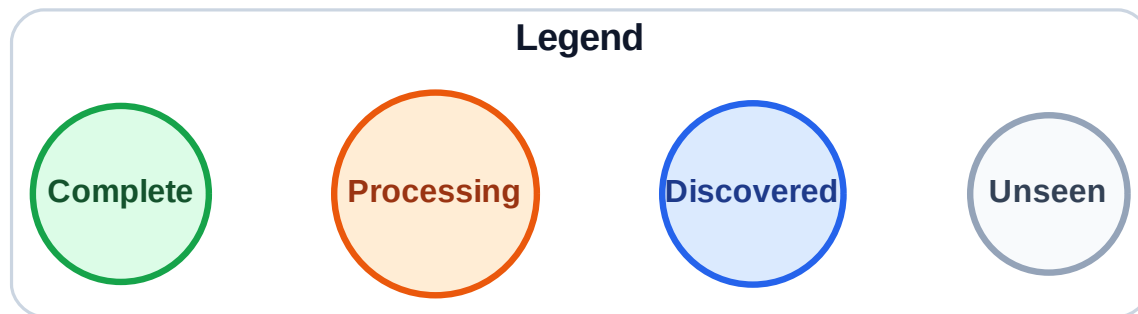
Stack (top at right): E | O | N | G | L | J | M

Visited: A, B, C, D, F, H



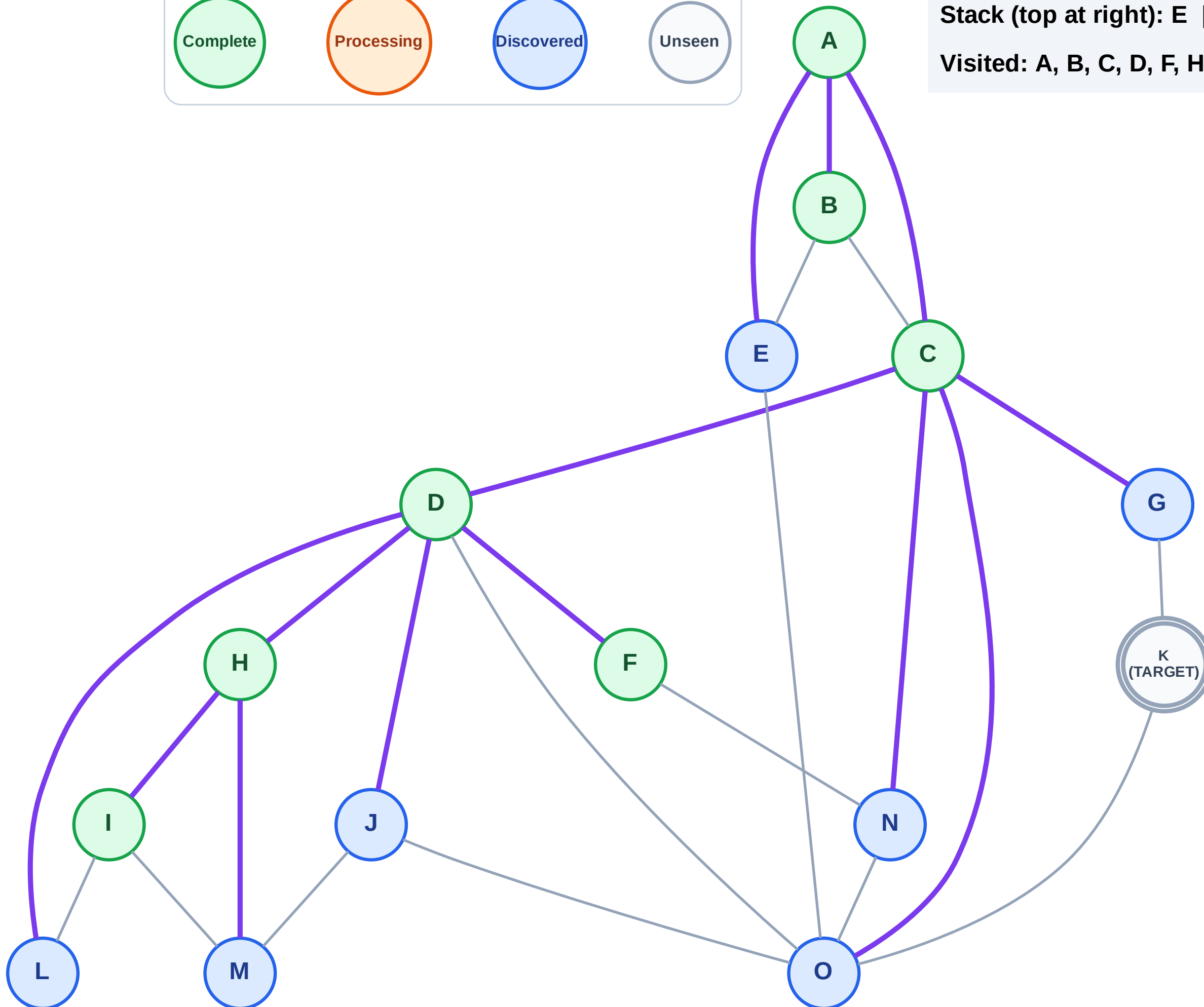
Step 15: No new neighbors from I

Step 15: No new neighbors from I



Stack (top at right): E | O | N | G | L | J | M

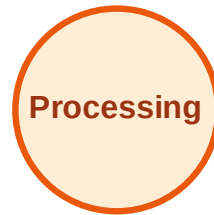
Visited: A, B, C, D, F, H, I



DFS Traversal

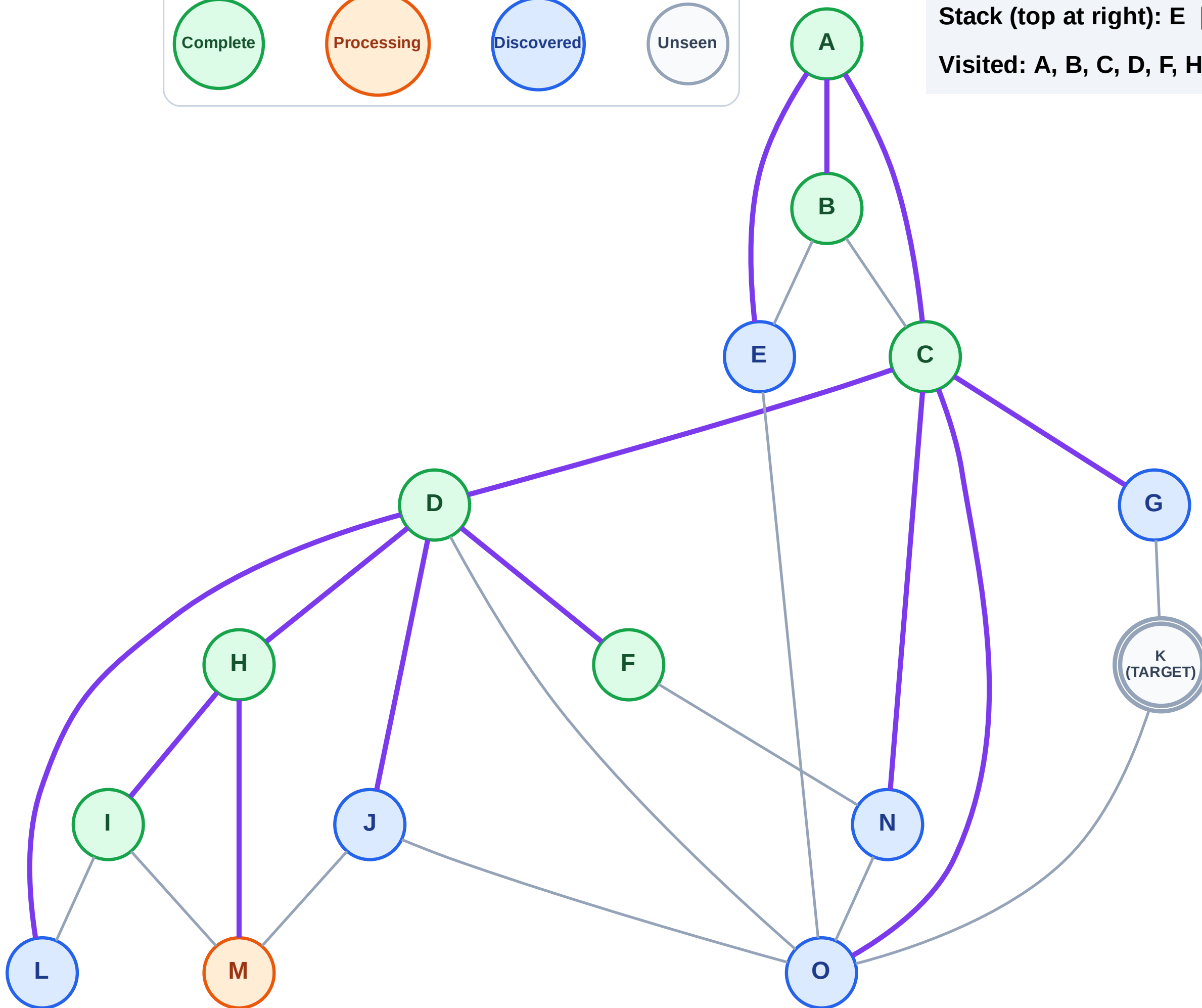
Step 16: Process node M

Legend



Stack (top at right): E | O | N | G | L | J

Visited: A, B, C, D, F, H, I



DFS Traversal

Step 17: No new neighbors from M

Legend

Complete

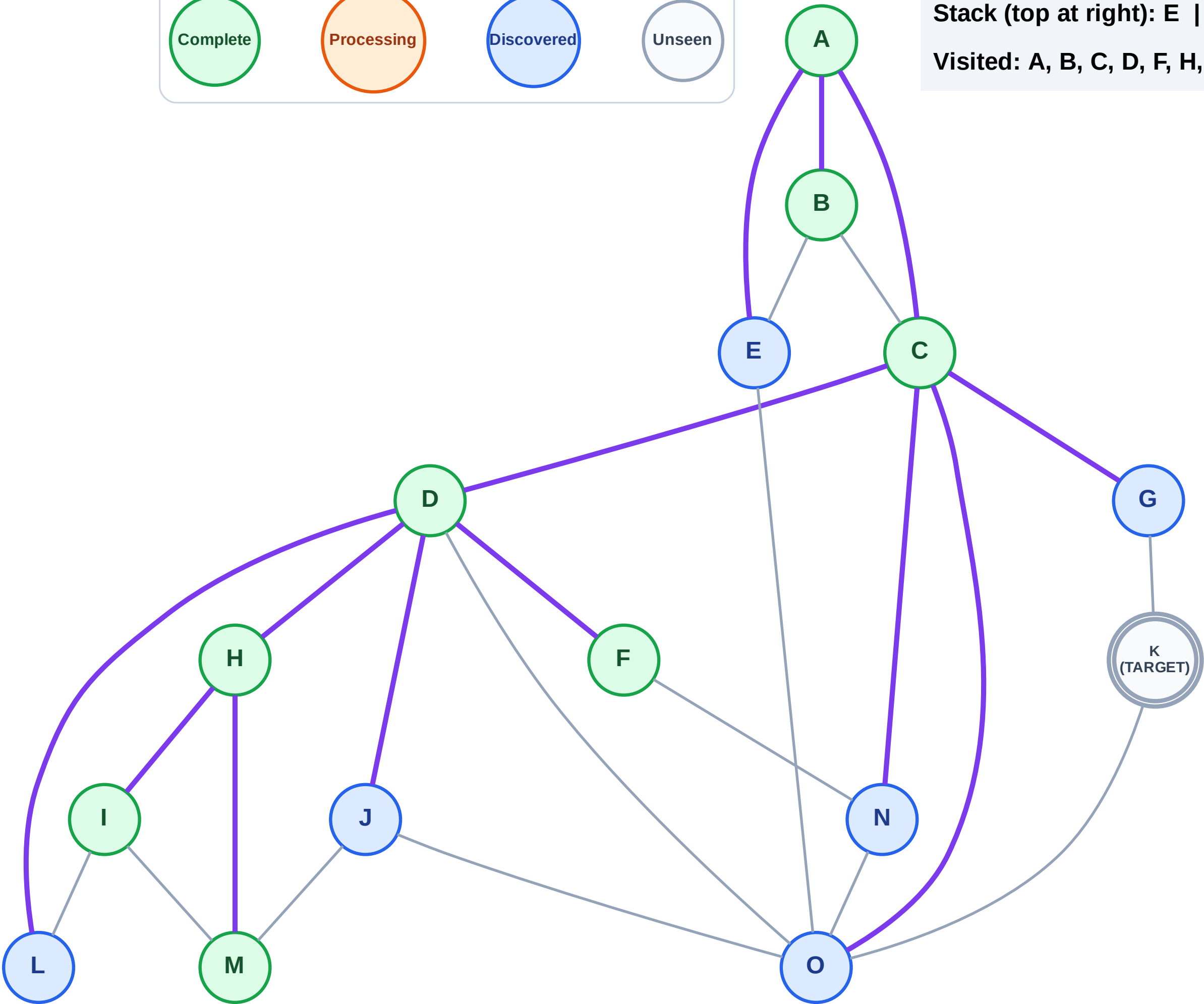
Processing

Discovered

Unseen

Stack (top at right): E | O | N | G | L | J

Visited: A, B, C, D, F, H, I, M



DFS Traversal

Step 18: Process node J

Legend

Complete

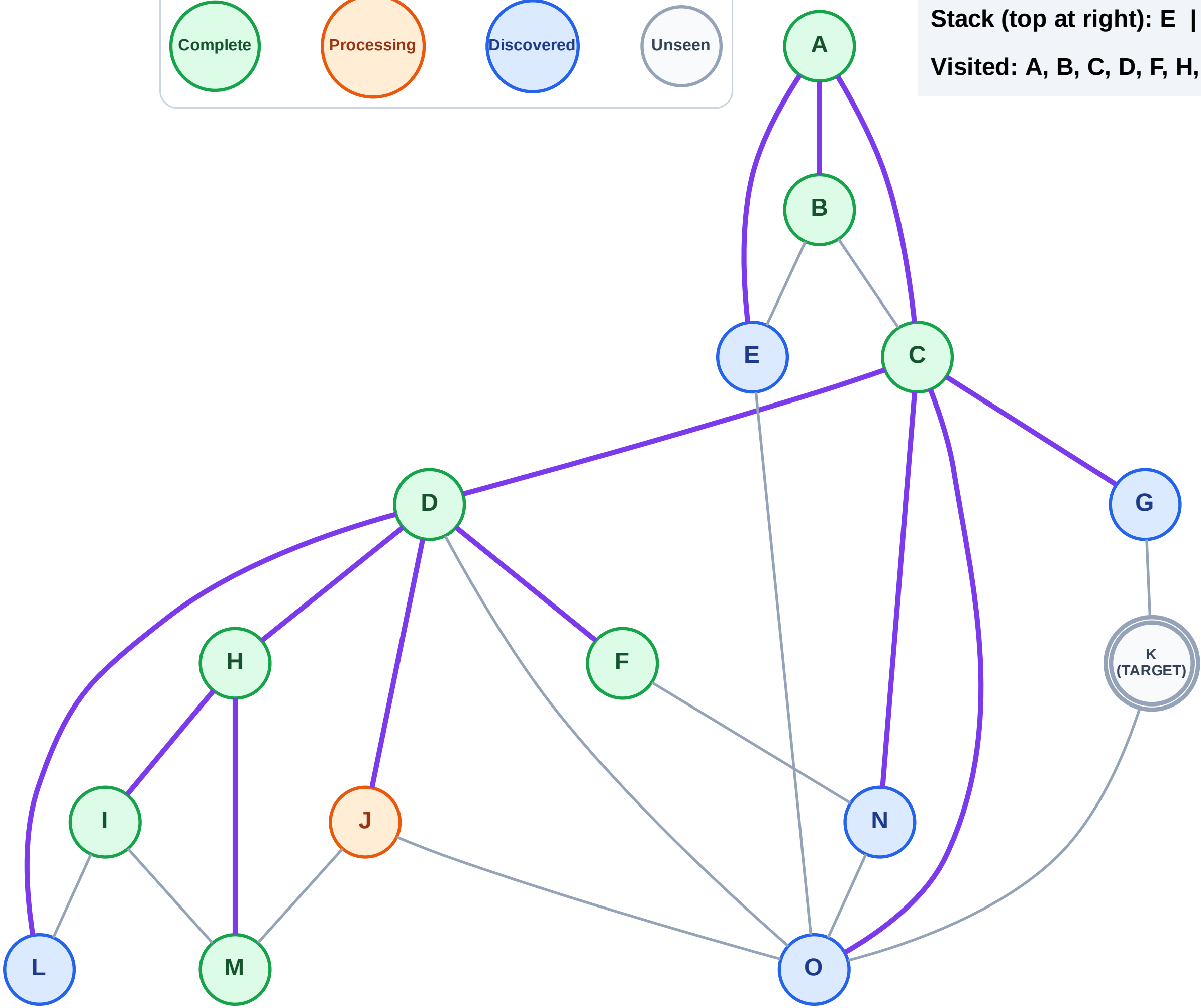
Processing

Discovered

Unseen

Stack (top at right): E | O | N | G | L

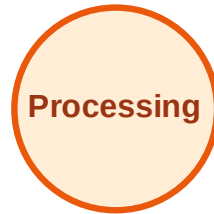
Visited: A, B, C, D, F, H, I, M



DFS Traversal

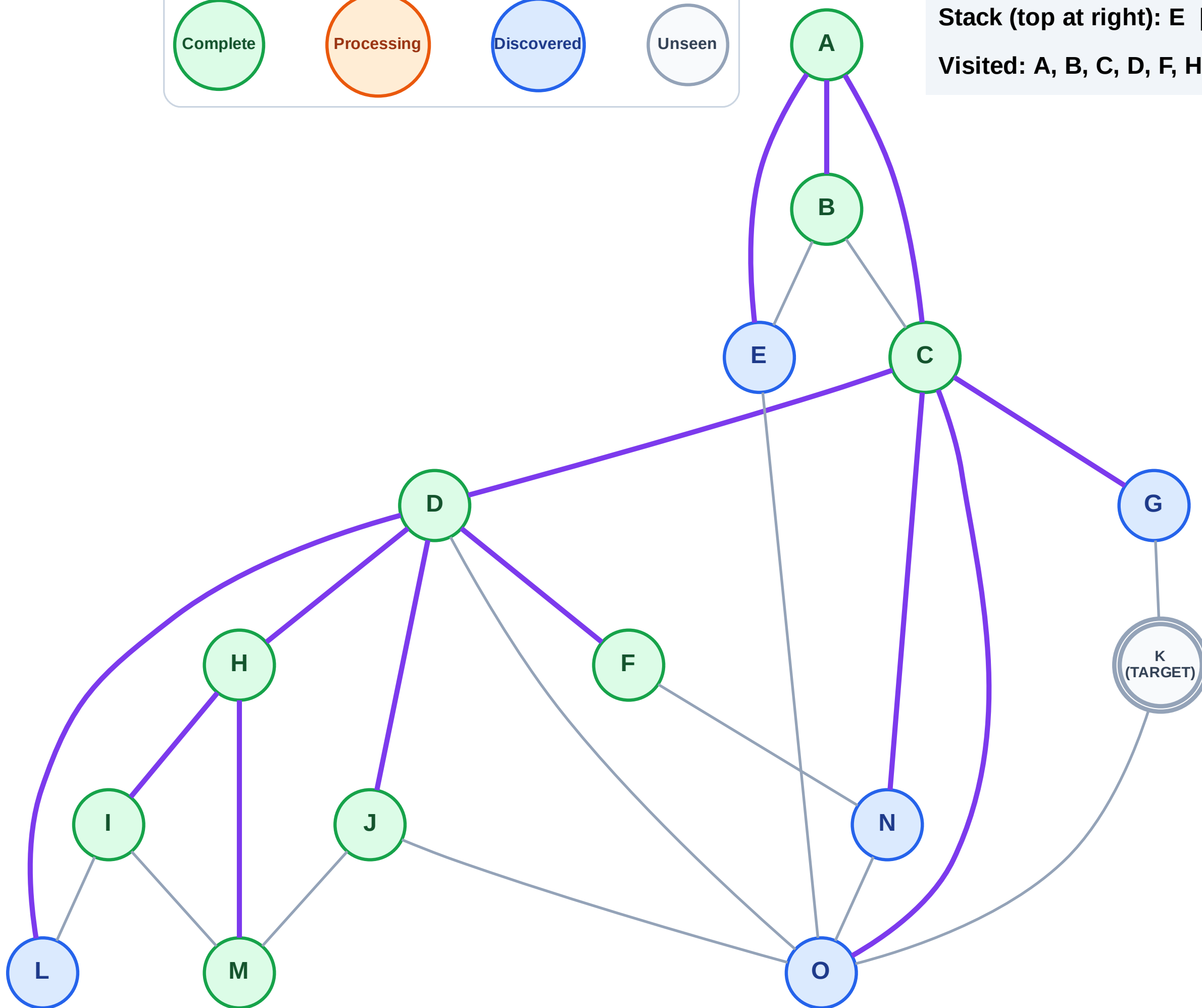
Step 19: No new neighbors from J

Legend



Stack (top at right): E | O | N | G | L

Visited: A, B, C, D, F, H, I, J, M



DFS Traversal

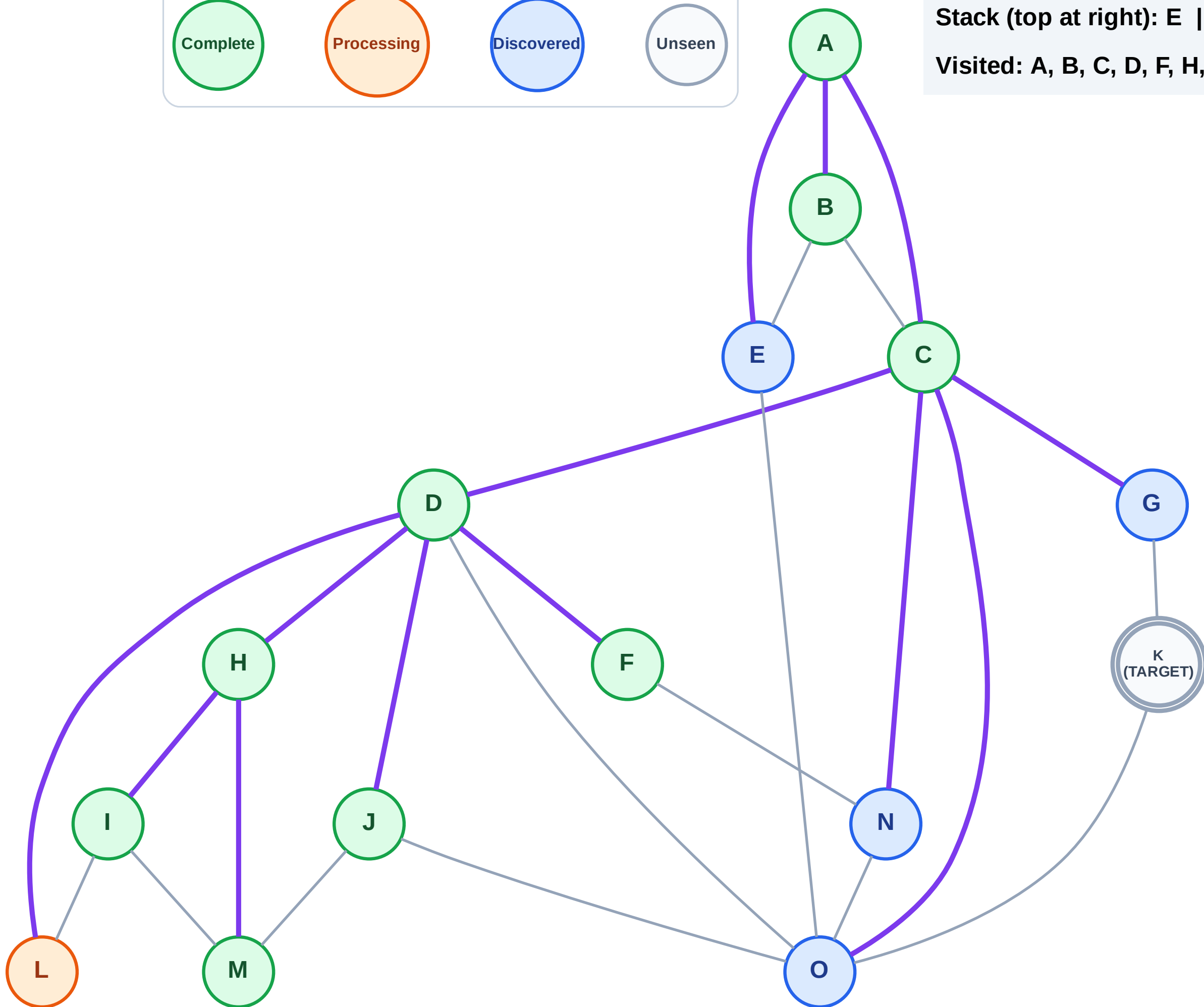
Step 20: Process node L

Legend



Stack (top at right): E | O | N | G

Visited: A, B, C, D, F, H, I, J, M



DFS Traversal

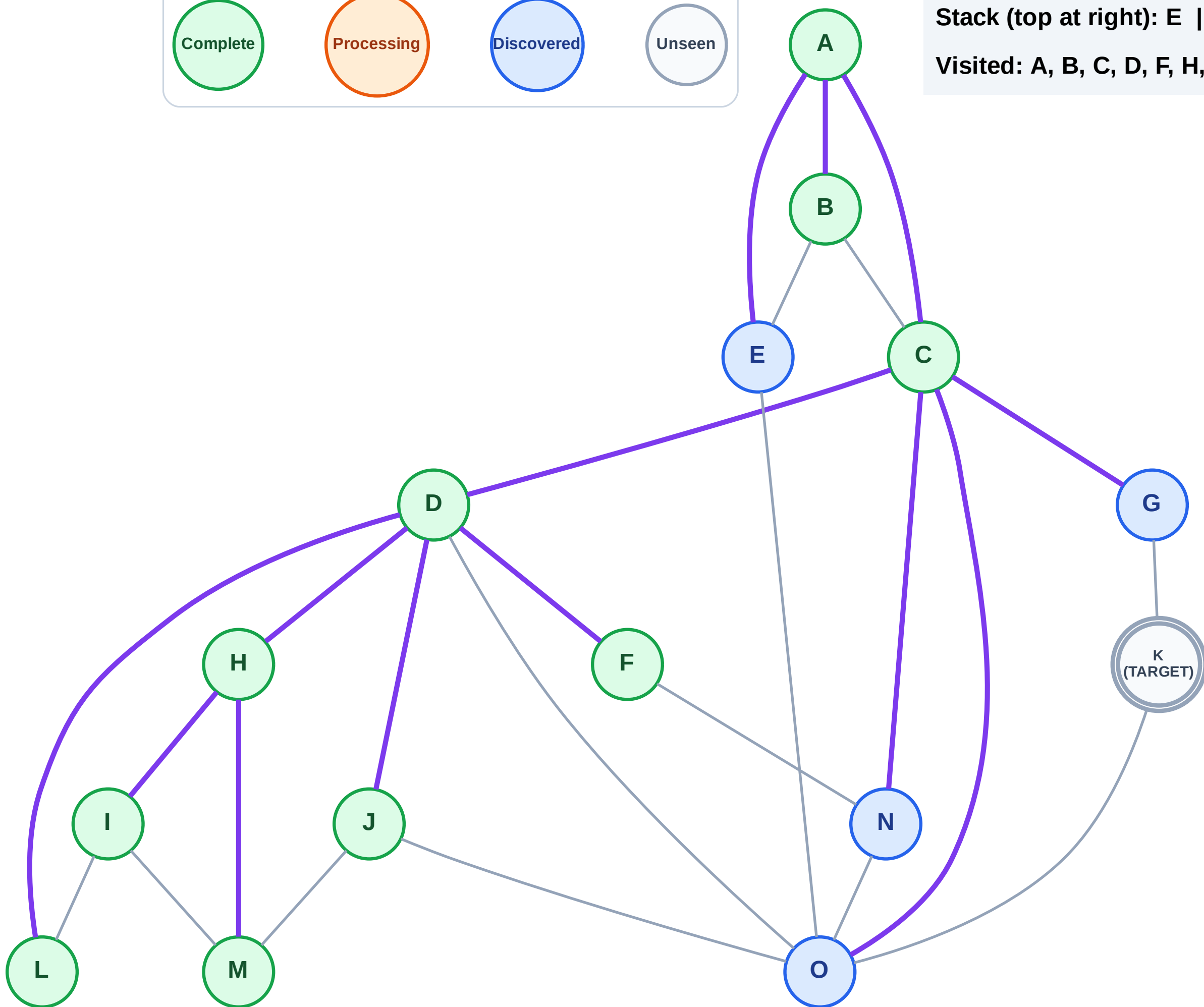
Step 21: No new neighbors from L

Legend



Stack (top at right): E | O | N | G

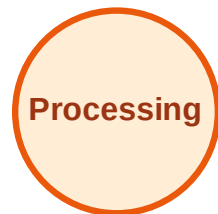
Visited: A, B, C, D, F, H, I, J, L, M



DFS Traversal

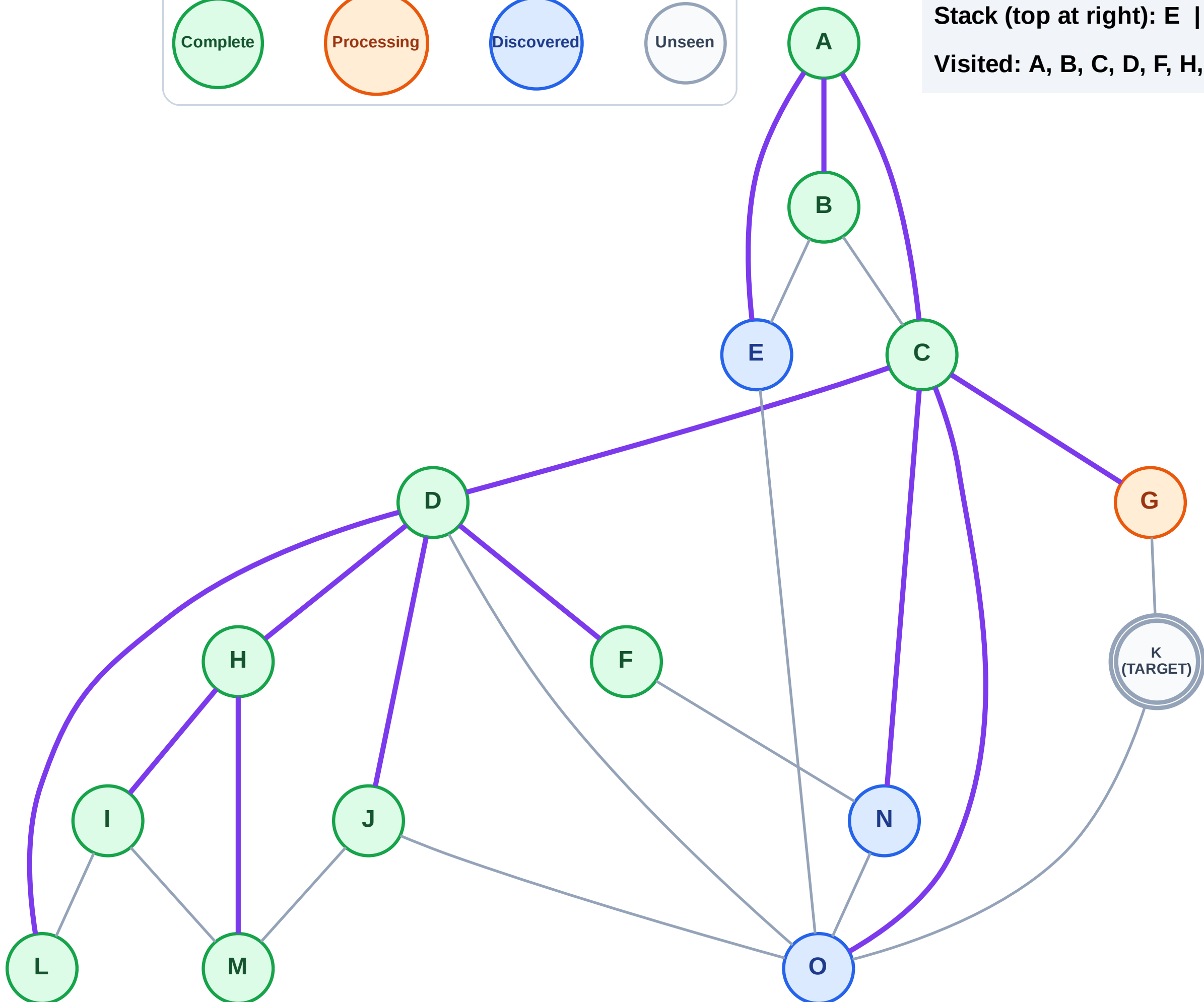
Step 22: Process node G

Legend



Stack (top at right): E | O | N

Visited: A, B, C, D, F, H, I, J, L, M



DFS Traversal

Step 23: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → C → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): E | O | N

Visited: A, B, C, D, F, G, H, I, J, K, L, M

